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Synchrotron Radiation Based X-ray Absorption Spectroscopy: Fundamentals and Applications in Photocatalysis

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Photocatalysis is one of the most promising green technologies to utilize solar energy for clean energy achievement and environmental governance. There is a knotty problem to rational designing high-performance photocatalyst, which largely depends on an in-depth insight into their structure-activity relationships and complex photocatalytic reaction mechanisms. Synchrotron radiation based X-ray absorption spectroscopy (XAS) is an important characterization method for photocatlayst to offer the element-specific key geometric and electronic structural information at the atomic level, on this basis, time-resolved XAS technique has a huge impact on mechanistic understanding of photochemical reaction owing to their powerful ability to probe, in real-time, the electronic and geometric structures evolution within photocatalysis reactions. This review will focus on the fundamentals of XAS and their applications in photocatalysis. The detailed applications obtained from XAS is described through the following aspects: 1) identifying local structure of photocatalyst; 2) uncovering in situ structure and chemical state evolution during photocatalysis; 3) revealing the photoexcited process. We will provide an in depth understanding on how the XAS method can guide the rational design of highly efficient photocatalyst. Finally, a systematic summary of XAS and related significance is made and the research perspectives are suggested.

1. Introduction

To confront the growing consumption of fossil fuels and environmental crisis, it is of particular urgency to explore the promising routes for renewable solar energy utilization.[1] Photocatalysis has become a realistic method with significant application in energy and environmental protection processes to develop new energy source and realize the carbon neutrality.[2] However, most photocatalytic applications was still difficult to realize industrialization due to their low photocatalytic efficiency.[3] Therefore, the rational design and development of high-efficiency photocatalyst is a critical issue.[4] Although much efforts have been devoted to develop efficient photocatalyst, such as the heteroatoms doping, cocatalyst loading, and hetrojuctions construction. insights into photocatalytic reaction mechanisms at atomic level or in spatial-temporal scale, including the active sites, local structure, and charge dynamics of photocatalyst are extremely challenging.[5] In most cases, the cocatalyst emerge with amorphous features or heteroatoms with ultra-low doping,[6] whose fine atomic structures are difficult to be deciphered based on conventional analytical instruments, such as XRD, SEM and XPS due to their lower sensitivity.[7] Meanwhile, the photocatalytic process is a multi-temporal spatial scale, the rapid dynamic evolution, structure transformation and reaction mechanism of photocatalyst are difficult to be explored by the conventional characterization techniques. Consequently, an in-depth understanding of the above processes is of significance in rationally designing photocatalyst to enhance photocatalytic activity and improve solar energy utilization. Many conventional characterization techniques can provide information about crystal structure, morphology, chemical composition, optical property, photogenerated charges' separation and recombination, and reactive intermediate species.[8] The main challenge is that photocatalytic reactions are quite complex processes, involving multiple electron/hole transfers and multi-temporal spatial scale. Recently, benefitting from the development of in situ characterization techniques that monitor photocatalyst under actual- or near-actual- reaction conditions, our understanding of the photocatalyst design and the photocatalytic performance has been dramatically improved.

Fortunately, synchrotron radiation based X-ray absorption spectroscopy (XAS) as an effective characterization can provide the information of electronic and local structure simultaneously. Synchrotron radiation is an electromagnetic radiation emitted when relativistic charged particles are accelerated radially. This radiation is produced artificially in accelerators based on storage ring by using bending magnets, undulators and/or wigglers. Main characteristics of this radiation are polarization, and the presence of wavelengths over the entire electromagnetic spectrum. Its exceptional characteristics in terms of intensity, brilliance, spectral range, time structure and now also coherence pushed many experimental techniques to previously un-reachable limits, enabling the performance of experiments unbelievable only few years ago. In the contrast, the most common laboratory XAS sources in the hard X-ray energy range

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are, due to their relatively simple working principle the classic, though highly improved X-ray tubes using various anode materials. It is well known that X-ray tubes convert only about 1% of the energy applied to the anode into radiation, while the majority of 99% is lost due to the conversion into thermal energy, therefore, the light of X-ray was lower and discrete than that of Synchrotron radiation. Synchrotron radiation has a series of excellent characteristics such as high brightness, continuous spectrum, wide spectral range, linearly polarized light, and high collimation.[9] And XAS is an element-specific, short-range, core-level spectroscopic method which can probe the local structure and electronic structure of particular atomic sites in chemistry, materials, and energy fields.[10] The main advantages of XAS over other commonly used characterization tools can be briefly outlined as follows: (i) XAS is a local probe. It can break the limitations of long-range order as required by XRD to investigate the local structure of nanoparticles or amorphous materials, even in the ultra-low concentration of few ppm.[11] Therefore, XAS is used to detect trace elements analysis in photocatalyst;[12] (ii) XAS probes the electronic nature of the absorber atom, providing details about the valence state and density of localized states close to the Fermi level;[13] (iii) XAS can tested a variety of materials not only in a static state but also in a dynamic state (in situ/operando conditions) without the demand of ultrahigh vacuum condition required by XPS and TEM.[14] Modern synchrotron light sources can provide a high X-ray flux with high time resolution down to the picosecond scale, which is applicable to the in situ photocatalytic reactions to detect the local structures evolution, photoexcited process and carriers dynamics of reaction process, providing the photocatalytic mechanism of photocatalyst.[15]

The photocatalytic process are involved at different timescales (Figure 1):[16] (1) photon excitation in femtosecond scale;[17] (2) charge carriers separation in femto- to picosecond scale;[18] (3) carriers diffusion/transport in nano- to microsecond scale,[18b] (4) the reactants activation/conversion in microseconds to milliseconds scale),[19] and (5) mass transfer/diffusion in milliseconds to seconds. The identification of these steps using the appropriate technique, can provide the guidelines to develop the "photocatalyst by design" concept and improve the overall efficiency.[20]

According to the abovementioned problem, in situ time-resolved XAS technique can effectively probe the charge transfer, dynamics chemical environment and oxidation state of each individual element in photocatalyst, helping us to unravel the local structures, geometric configurations, charge transfer, and electronic structural evolution of photocatalyst.[9b][21] In particular, with the advent of fourth generation synchrotron sources, in situ time-resolved XAS with femtosecond-nanosecond time resolution allows for the following of real time evolution in the electronic and geometric structure of a photocatalyst under operating conditions.[22] The in situ or operando XAS has evolved into a powerful tool to the dynamic electronic and coordination structure of photocatalyst under realistic reaction conditions as a basis for structure activity relationships.[23] In situ X-ray absorption near-edge structure (XANES) could be widely used to monitor variations in the electronic state of the selected atomic components, while in situ extended X-ray absorption fine-structure (EXAFS) can be applied to gauge the variations in the local symmetry, phase composition, and electronic properties of the targeted atom during catalytic processes.[24] These in situ XAS tools are particularly applicable to reveal the photoexcited process, the evolution of active sites under reaction conditions, and charge transfer process.[25] With this regard, the present review first provide the brief description about the fundamentals of XAS. In addition to that, the development of XAS will also be presented. Furthermore, the applications of XAS in photocatalysis including



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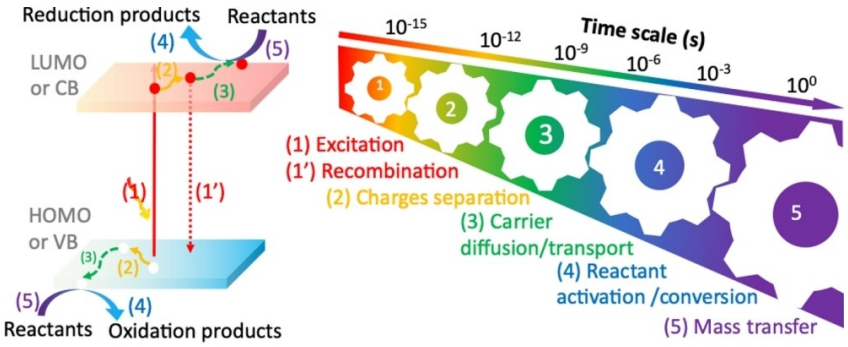


Figure 1. Illustration the photocatalysis process in different time scales. Reproduced with permission from Ref. [22].

the local structure of photocatalyst, the dynamic structure and chemical state of photocatalyst, the charge transfer and the photoexcited process will be reviewed. Finally, it ends with a summary of current challenges and outlook for future research. This review will provide more guides for the fabrication of highly-performance photocatalyst, and meanwhile promote the utilization of XAS techniques in the field of photocatalysis.

2. The Fundamentals of Synchrotron Radiation X-Ray Absorption Apectroscopy

2.1. Basic Principles of Synchrotron Radiation X-Ray Absorption Spectroscopy

Synchrotron radiation is an electromagnetic radiation emitted when relativistic charged particles are accelerated radially. This radiation is produced artificially in accelerators based on storage ring by using bending magnets, undulators and/or wigglers (Figure 2). Main characteristics of this radiation are polarization, and the presence of wavelengths over the entire electromagnetic spectrum. Synchrotron radiations have unique properties because of their high brilliance, collimations, broad energy spectrum, and coherence power that break the limits to characterize the material properties.[26] Synchrotron radiations sources have been continuously updated to improve resolution and minimum sample quantity required for measurements.[27]

According to its energy level, X-rays can be divided into hard X-rays (photon energy above 5–10 keV), and soft X-rays (photon energy below 5 keV).[28] Hard X-ray spectroscopy (> 5 keV) is used to study transitions from and to the 1 s shell (K-edge) and valence electron orbitals of transition metals involved in catalysis to uncover the geometric and electronic structure of the metal centers. Soft X-ray spectroscopy (< 1 keV), transitions from and to the 2p shell (L-edge) of transition metals to the valence orbitals, is used to probe the charge and spin distribution and the degree of covalency of bonds, all of which are critical properties for transition-metal-based catalysis. The hard X-rays cover the K-edge region of heavy elements (such as transition metals elements Cd, Ti, Pt, Pd, and Ag),[29] which commonly used in photocatalysts such as CdS, TiO_2, $SrTiO_3$,

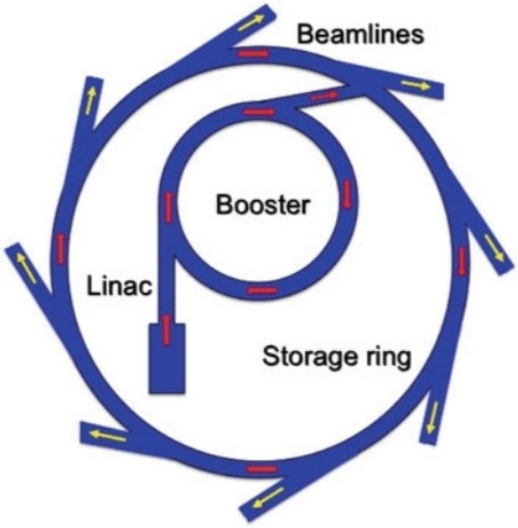


Figure 2. Schematic planar view of a synchrotron radiation facility. Reproduced with permission from Ref. [26a]

et al.,[30] and cocatalysts (such as Pt, Pd and Ag et al.,)[31] Compared with hard XAS, the soft XAS is used to measure the K-edge region of light elements (such as B, C, S, N, and O) and the L-edge of transition metal (for example Co, Ni, Cd, Ti, and Pt), which are all relevant for photocatalysts (such as g-C_3N_4, CdS and TiO_2)[32] and cocatalysts (such as CoOOH, Ni_2P, and MnO_2).[33] The K and L edges X-ray absorption spectroscopy (XAS) were intrinsically corresponding to the excitations of the core electrons from *s* and *p* orbits to the *p* and *d* orbits, respectively.[34] L-edge spectra often show more structure dependence on redox state, hence the use of L-edge over hard X-ray K-edges could be justifiable in this case. The Ti L-edge is especially sensitive to oxidation state and least squares fitting to reference compounds can readily be used to estimate the extent of oxidation. The oxidation state sensitivity and structural detail of the non-1 s level transitions such as 3rd row transition metal L-edges is often greater than their K-edge counterparts. Therefore, as for the 3d transition metals, the information obtained through hard X-ray and soft X-ray absorption spectroscopy techniques are often complementary to explore the local structure of materials.[35]

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XAS is a technique monitoring the X-ray absorption coefficient of a sample as a function of incident X-ray photon energy (E) below and above the absorption edge of selected atoms.[6] As each element has its characteristic absorption edges, XAS can be used for qualitative analysis of elements. Figure 3a displays an example of Pd K-edge XAS spectrum of Pd foil and different edges that arise from corresponding electronic transitions. Excitation of electrons from 1 s, 2 s, $2p_{1/2}$, and $2p_{3/2}$ corresponds to the K, L_1, L_2, and L_3 edges, respectively. Inhere, the K-edge XAS spectrum in Figure 3a arises due to the excitation of orbitals from the core (1 s orbital) to unoccupied orbitals and continuum. The XAS provides the information on the local structure and chemical states of the atomic cluster in a short range, regardless of whether the sample is crystalline or non-crystalline.[37]

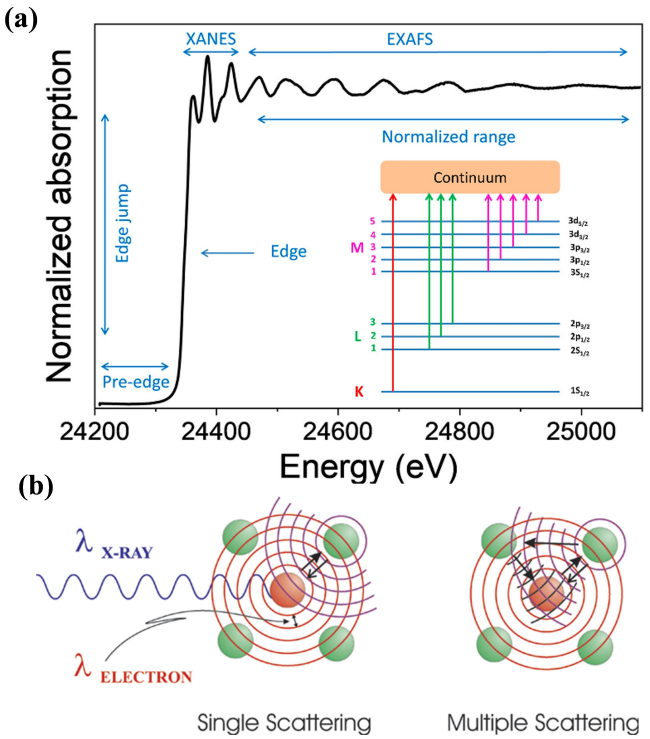


Figure 3. (a) Schematic representation of XAS spectra of Pd foil at Pd K-edge with various features and electronic transitions giving rise to different absorption edges and (b) Pattern of an outgoing and backscattered photoelectron wave in the case of EXAFS (single scattering events) and XANES (multiple scattering events). Reproduced with permission from Ref. [26].

The absorption edge energy corresponds to the energy required to eject an electron from an electron shell of an element of interest.[38] In general, the XAS spectrum can be observed two categories structure information: X-ray absorption near edge structure (XANES) and extended X-ray absorption fine structure (EXAFS). The important features of XANES is pre-edge and edge, its region is the absorption edge and ~50 eV above it. XANES is sensitive to local atomic states such as oxidation states, site symmetry and local electronic structure of the absorbing atom, which is caused by the multiple scattering of the excited photoelectron, as shown in the right of Figure 3b. It can observe the information about the oxidation states and the d-band characteristics, further provide the structural information including orbital configurations and symmetry. Generally, as the valence state increases, the shielding effect weakens, orbital energy decrease, and absorption edge blue shift. As for EXAFS, the absorption energy region extends from 50 to 1000 eV above the absorption edge, which generated mainly by the single scattering of absorbing atoms (the left of Figure 3b).[39] The data of EXAFS in energy space can be converted that in wavelength k-space and R-space.[40] These results can give the information about the coordination structural information, including the coordination number and bond length.[41] In general, the information observed from XANES and EXAFS are highly complementary.[42] Figure 4 shows the W L_3-edge spectrum of WO_3, demonstrating the features of a typical XAS spectrum. The XAS spectrum of WO_3 at the W L_3 edge consist of two parts. The XANES structure can provide the information of the valence states of W^{6+} with the feature of absorption edge, which caused by the orbital transition from the W $2p_{3/2}$ to 5d orbitals.[43] EXAFS region appears because of the oscillations of the outgoing electron by surrounding atoms, which offers information regarding local coordination of W, including bond lengths, bond angles and coordination number of W–O bond.[35a,44]

Methods of measurement depend on whether transmitted X-rays, fluorescent X-rays, or photoelectrons are monitored when the sample is irradiated with incident X-rays (Figure. 5).[46] In the transmission mode, the difference between the incident

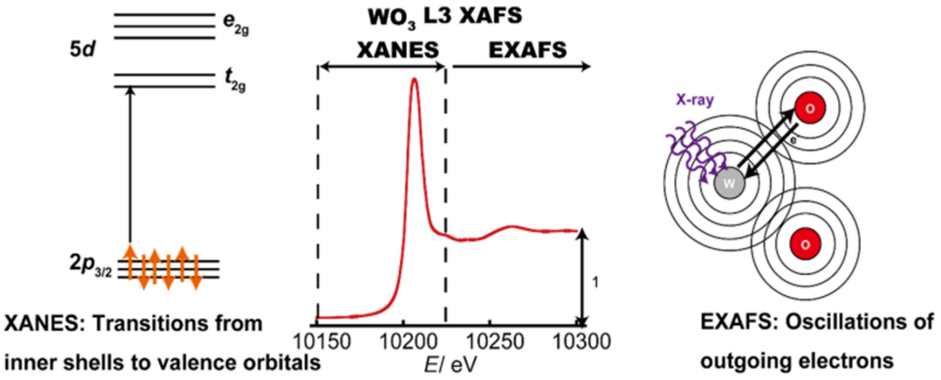


Figure 4. Schematic illustration of typical W L_3-edge of WO_3

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and directly transmitted X-ray intensity are measured (Figure 5a). The transmission mode is the most convenient for collecting XAFS data.[47] In this mode, the spectrum is obtained by measuring the intensity of the radiation before and after the sample as a function of the energy of the incident radiation, the X-ray absorption coefficient (μ(E)) depends on the photon energy *E* and the sample of thickness *d*. The transmitted intensity I_t could be calculated according to Beer's law ($I_t = I_0 e^{-\mu(E)d}$). Transmission mode XAFS is mostly suitable for concentrated samples in which the element of interest is the major component (>10%).[48] XAFS can be also observed from the fluorescent X-rays emitted by X-ray absorption of specified atoms, the fluorescence mode by measuring the outgoing fluorescence signal is widely used for the dilute samples and for surface adsorption situations. The sensitivity of fluorescence mode measurement is extremely high when compared with transmission mode measurements.[49] Fluorescence mode was shown schematically in Figure 5b. The intensity of this fluorescence is proportional to the absorption efficient caused by the investigated element. Thus, it is good for dilute samples, like the noble metal cocatalyst or heteroatoms doping. Because of experimental constraints and limited penetration depth of low energy photons or electrons, the fluorescence mode (with a backscattering geometry) is most convenient for in situ/operando experiments. Instead of measuring the emitted fluorescence, XAFS can be observed by measuring the total yield of Auger electrons and secondary electrons emitted following absorption of X-rays by atoms (Figure 5c).

2.2. The Development of X-Ray Absorption Spectroscopy

X-ray absorption fine structure (XAFS) spectroscopy, also named X-ray absorption spectroscopy (XAS), is a well-established and widely used characterization of materials in solid or liquid, crystalline phase and amorphous phase. Changes of structure and chemical states of catalysts have been successfully observed with XAFS under reaction conditions. XAFS spectroscopies have advanced rapidly since the development of synchrotron radiation facilities that provide intense X-ray radiation with a continuous spectrum that can reach 120 keV.[50] With the advent of extremely high brilliant sources, time-resolved XAS studies on highly diluted materials systems become feasible with unprecedent signal-to-noise (S/N) ratios.[51] Among the time resolved XAS methods, the XAFS spectra can be obtained within several milliseconds by using quick scan XAFS (QXAFS)[52] and energy dispersive XAFS (DXAFS).[53] The QXAFS measurement was developed by Frahm et al. in 1988 and has since been improved upon and used worldwide in synchrotron facilities as a standard XAFS method in the analysis of dynamic events such as catalysis. Figure 6a shows the schematic of QXAFS for transmission mode. When Q-XAFS is performed combined with transient response techniques, the changes in the structure of the catalysts and the structure-function relationship at the electronic structure level can be detected in real-time.[54] The development of energy dispersive XAFS (DXAFS) has allowed for further progress in the QXAFS technique, achieving temporal resolutions down to the microseconds (μs) scale. Instead of using a monochromator, DXAFS utilizes a curved crystal, known as a "polychromator", to disperse the X-ray beam in space. This dispersion causes photons with different energies to travel along different paths, meanwhile, the acquisition scheme of absorption spectrum in all photons energy is parallel. After passing through the sample, the transmitted beam is detected by a position-sensitive detector, which transforms the energy-direction correlation into an energy-position correlation. Therefore, the EDXAS spectrometer could operates in a time resolved manner. Besides the acquisition speed, another major advantage of EDXAS is the stable energy scale and focal spot position.

Since photocatalytic reaction including the photon excitation of semiconductor, charge carriers separation and recombination, transfer process and surface photocatalytic reaction, happened in a very fast time-scale, even in the femto to picosecond scale. In order to observe such fast events, the demanding time resolution of XAFS measurements is much

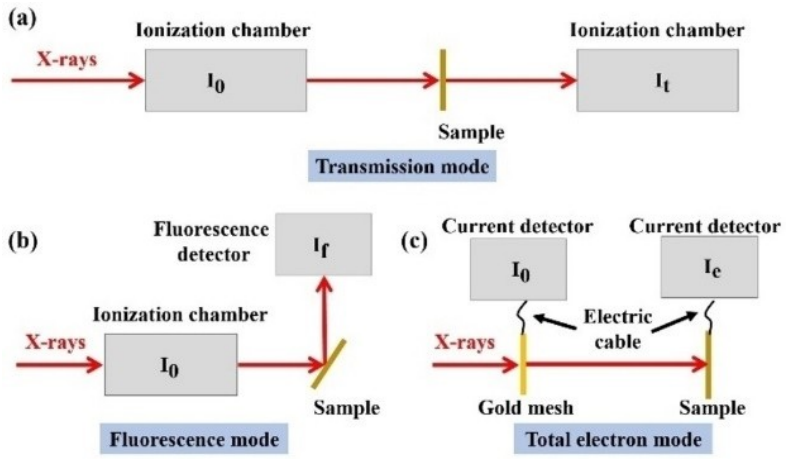


Figure 5. Schematic illustration of the XAS experiment setup in (a) transmission mode, (b) fluorescence mode, and (c) electron yield mode. Reproduced with permission from Ref. [45].

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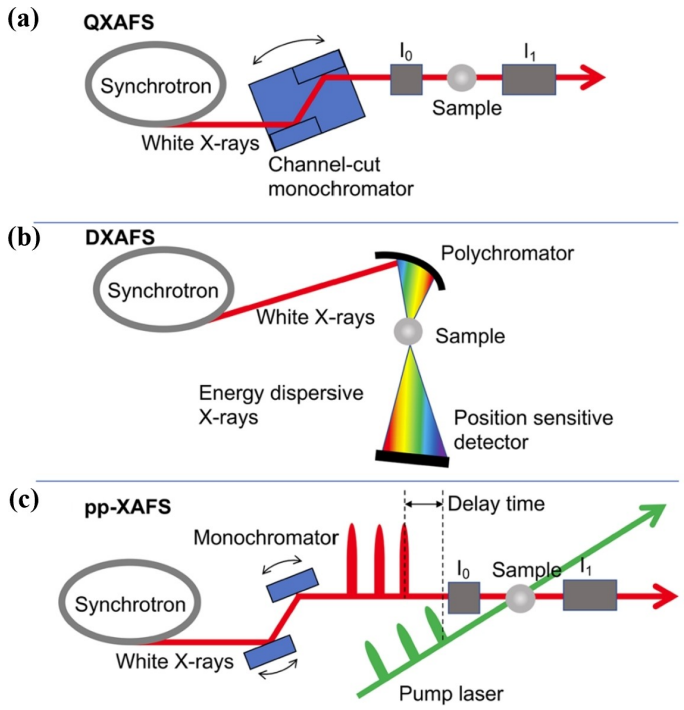


Figure 6. Schematics of (a) Quick-XAFS (QXAFS), (b) Energy-dispersive XAFS (DXAFS), and (c) pump-probe XAFS (pp-XAFS). Reproduced with permission from Ref. [59].

shorter than the time resolution of QXAFS and DXAFS techniques. According to this issue, the pump-probe XAS technique is applicable to analyze ultrafast and repeatable processes such as the measurements of photogenerated charge carriers' lifetimes and charge transfer process of catalytically active species in a time resolution of nanoseconds or less.[57] As shown in Figure 6c, X-rays coming from synchrotrons facilities are short pulses with finite pulse width (typically about 100 picoseconds (ps)).[58] The time resolution of the pump-probe XAFS measurements is determined by the duration of the X-ray pulse of synchrotron facilities, which is the order of around 100 ps.[59–60] Furthermore, the high peak intensity of synchrotron facilities allows for the application of XAFS techniques like high energy resolution fluorescence detection X-ray absorption spectroscopy (HERFD-XAFS), which offer high energy resolution fluorescence detection. However, the introduction of X-ray free electron lasers (XFELs) with very short pulse durations has expanded the possibilities of pump-probe XAFS measurements to the femtosecond range.[61] The time-resolved XAS technique with femtosecond timescale has significantly influenced the comprehension of photochemical reactions by allowing real-time investigation of the local electronic and coordination environments of photocatalyst, thereby further providing a comprehensive understanding of the photocatalytic mechanism.

As mentioned above, XAS, which includes XANES and EXAFS. XANES can reflect the detailed electronic states information about the oxidation state, site symmetry, and chemical bonding of the given absorbing atoms.[62] EXAFS oscillations can reflect the local structure information including coordination bond length, coordination numbers, and coordination atom species for the special absorbing atom. However, since photocatalytic processes occur on the sub-microsecond timescale, time-resolved XAS studies with high resolutions was more useful for probing the local structure evolution of photocatalyst during the photocatalytic processes and elucidating charge transfer processes. Furthermore, the behavior of photocatalyst during the reaction remains a black box under these ex-situ characterizations.[63] The utilization of in situ, time-resolved synchrotron XAS techniques offers unique opportunities to study the dynamic structure evolution of working (in situ/operando) catalyst to clarify the photocatalytic reaction mechanism.[64] Over the past few years, in situ XAS experiments have been successfully employed in photocatalyst to explore the dynamic local structure evolution of photocatalyst upon reaction conditions, while in some cases, it allows retrieval of valuable information about their structural evolution. Especially, XANES spectrum is suitable for time-resolved experiments due to its short collection time, sensitivity to chemical information such as electron states not occupied by valence states and charge transfer, and weak dependence on temperature, and can be used for in-situ chemical experiments.[65] By using in situ time-resolved XAS, the direct determination of the photoexcited process, the photogenerated charge transfer, the chemical state and local structure evolution of photocatalyst during photocatalytic process is made possible, allowing clarification of the reactions mechanisms of photocatalysis process.[12a,66] Furthermore, time-resolved XAS conducted in situ or operando is a valuable tool to monitor the dynamic changes (chemical states changes, coordination structure and charge transfer behavior) of photocatalyst in real time.[21b,67] By analyzing the atomic structure and electronic state of the active site, as well as the charge transfer behavior, the relationship between structure and properties is established, which provides a new perspective for designing highly efficient photocatalyst and investigating their photocatalytic reaction mechanism.[68]

3. Application of X-Ray Absorption Spectroscopy in Photocatalysis

To characterize such complex and dynamic photocatalytic systems, X-ray absorption spectroscopy (XAS) has been shown to be a very valuable tool since it provides element-specific information of the local structure of a special atom, further suggesting the active sites, interfacial interaction, photoexcitation process, and charge carriers dynamics.[69] The in situ XAS experiments further reveal information about the dynamic local structure evolution and electronic structure evolution of photocatalyst during reactions, allowing clarification of the reactions mechanisms of photocatalysis process.[70] In this section, we summarize the applications of XAS characterization techniques for direct monitoring of the electronic, coordination structure and charge transfer behavior of photocatalysts.

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3.1. Uncovering the Local Structure of Photocatalysts

3.1.1. Ascertaining the Local Structures of the Cocatalysts

Although heterogeneous semiconductor-based photocatalysts have been extensively studied, how local structural evolution contribute to critical properties and can be rational designed and regulated remain understudied because of the complex photocatalytic reaction process and complex photocatalysts material systems.[71] Without modification, blank semiconductors (such as $BiVO_4$,[72] TiO_2, GaN,[73] and CdS,[74] et al.) usually exhibit the low photocatalytic performances, which have turned out to be fatal for the industrialized application.[75] To address this issue, different strategies involving cocatalyst loading, heterojunction construction, and heteroatom incorporation are the effective strategies to improve the photocatalytic efficiency,[76] in which the cocatalyst loading has been demonstrated to the most effective routes, since cocatalyst can not only improve the separation and transfer efficiency of photogenerated electron-hole pairs, but also provide the more active sites to accelerate the photocatalytic reaction kinetics.[77] Therefore, the effective regulation and the in-depth understanding of the microscopic enhancement mechanism of cocatalyst is very critical to achieve further increases in photocatalytic activity. Inhere, clearly identifying the cocatalyst through investigating the structural characteristics of the cocatalyst will be highly beneficial for unraveling their working mechanisms in more detail. Many works have used XAS as a tool for providing the information of local and electronic structure of cocatalyst and illustrate the structure-activity relationship.

N-type monoclinic bismuth vanadate ($BiVO_4$) is a promising visible-light-responsive photocatalyst, cocatalyst loading is considered as an effective strategy to improve the photocatalytic activity of $BiVO_4$. The local and electric structure of cocatalyst are critical to the enhancement of the photocatalytic performance. Inhere, the electronic structure of $FeCoO_x$ cocatalyst in $FeCoO_x/BiVO_4$ photocatalyst were analyzed by Fe and Co K-edge XANES in Figure 7. From Figure 7a–b, it can be obtained that their oxidation states of both Fe and Co can be confirmed to be Fe^{3+} and Co^{3+}. From the R space of Fe K-edge and Co K-edge in Figure 7c, it can be observed that the Fe–O bond from $FeCoO_x/BiVO_4$ is short than Fe–O–Fe bond (2.86 Å) from FeOOH and Co–O bond from $FeCoO_x/BiVO_4$ is longer than Co–O–Co bond (2.69 Å) from CoOOH, which indicating the formation of FeCoOx composite on the surface of $BiVO_4$. Furthermore, the FT-EXAFS fitting curves of Fe K-edge and of Co K-edge in Figure 7d–e further verified the formation of $FeCoO_x$ phase, which not only lowers Gibbs free energy barrier of water oxidation, but also makes a better promotion on the electron transfer as well as charge separation, further improve the photocatalytic performance due to the optimized electronic structure of Co active sites due to the introduction of Fe atoms. The nanocomposite of FeOOH and CoOOH (denoted as $FeCoO_x$) in situ formed on the {110} facet of $BiVO_4$ not only lowers Gibbs free energy barrier of water oxidation, but also makes a better promotion on the electron transfer as well as charge separation compared with the commonly used CoO_x cocatalyst. Based on these results, the phase species of the $FeCoO_x$ on the surface of $BiVO_4$ sample can be deduced to be more similar to FeOOH/CoOOH with respect to Fe_2O_3/Co_2O_3. In addition, compared with the corresponding single-phase hydroxides FeOOH and CoOOH, $Fe/BiVO_4$ sample exhibits much shorter Fe–O bond and longer Co–O bond, and the length of Co–O–Fe bond is between Co–O–Co and Fe–O–Fe (Figure 2e). This demonstrates the existence of electron transfer and a strong interaction between Fe and Co in the $FeCoO_x/BiVO_4$ sample, providing further proof about the formation of the nanocomposite.[72]

Although noble metal such as Rh can act as an efficient cocatalyst to promote photocatalytic water splitting performance, it is also active for the reverse reaction ($H_2+O_2 \rightarrow H_2O$),

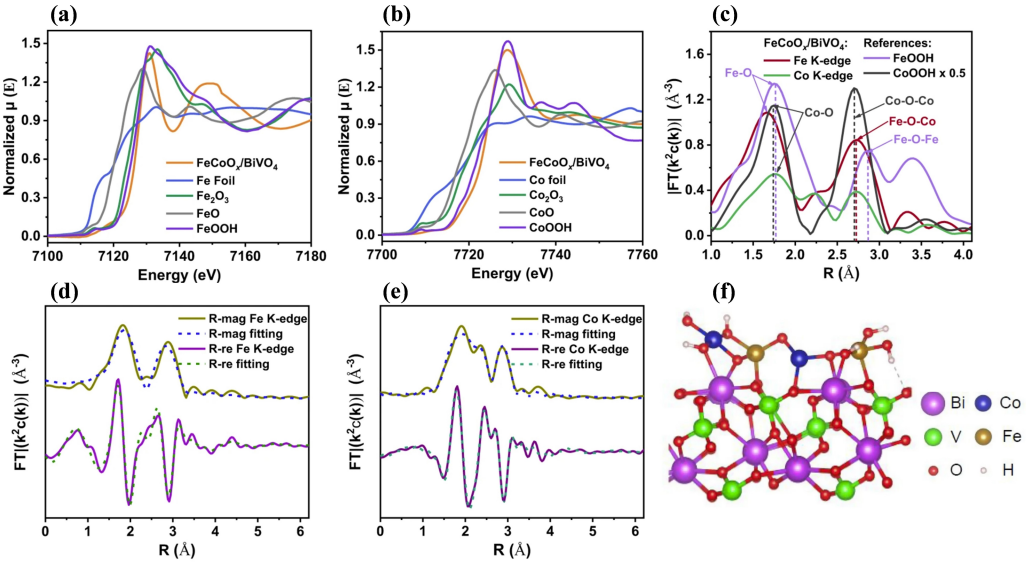


Figure 7. Normalized (a) Fe K-edge and (b) Co K-edge XANES spectra, (c) Fe K-edge, and Co K-edge, (d) Fe K-edge and (e) Co K-edge FT-EXAFS fitting curves at R space, (f) Schematic diagram of electric structure of $FeCoO_x/BiVO_4$ interface. Reproduced with permission from Ref. [72].

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which significantly limits its usefulness. To address this issue, Domen and co-workers designed and prepared a core/shell-structured cocatalyst (noble-metal/Cr_2O_3 nanoparticle) to improve the photocatalytic H_2 evolution activity because the Cr_2O_3 shell can effective prevent the occurrence of reverse reaction. The local and electronic structure of noble-metal/Cr_2O_3 cocatalyst was clarified through the use of XAS method. By comparing the XANES and EXAFS spectrum of Rh and Cr in Rh/Cr_2O_3-photodeposited ($Ga_{1-x}Zn_x$) ($N_{1-x}O_x$) with that in their corresponding reference materials (Figure 8). XANES was able to identify the valence states of Rh and Cr in the Rh/Cr_2O_3 (core/shell) nanoparticles, the results showed that the Rh species was metallic with the valence state of +0 and the Cr species was the valence state of +3 with the phase of Cr_2O_3, which was also verified by the EXAFS spectrum. The XAS results confirmed that the Rh^0 nanoparticles were covered by Cr_2O_3 shell. The Cr_2O_3 shell suppresses water formation from H_2 and O_2 on Rh nanoparticles, allowing stoichiometric decomposition of pure water to be achieved. The core/shell structure also provides enhanced H_2 evolution compared to that of bare Rh nanoparticles.[73]

Another example of XAS characterization to identify the cocatalyst CoP with single-atom phosphorus vacancies defects. As it can be seen from XANES spectra in Figure 9a, compared with Co foil, the blue shift in the absorption edge of the Co k-edge in CoP–V_p@$Cd_{0.5}Zn_{0.5}S$ (CoP@CZS)was observed, which suggested the increased electron density on Co atoms due to the single-atom phosphorous vacancies. Furthermore, the local structure of CoP–Vp in CoP–V_p@ CZS were investigated by EXAFS spectrum and wavelet transform plots in Figure 9b–c,

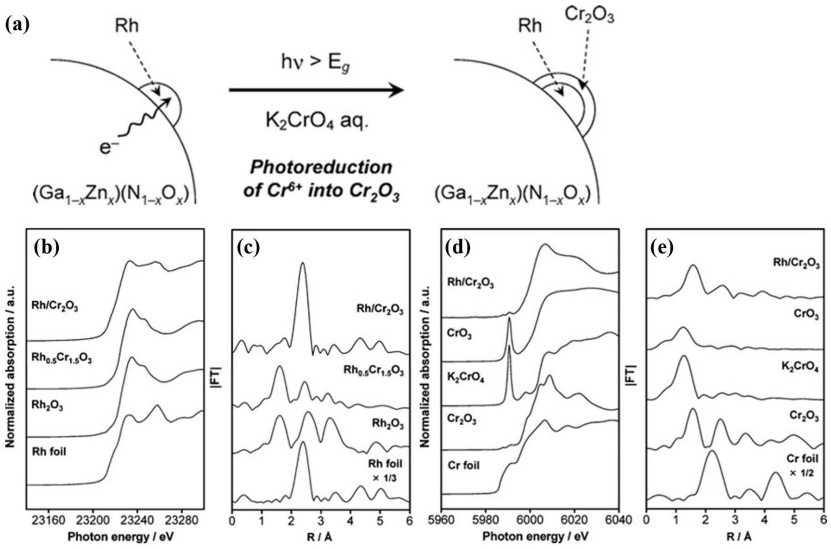


Figure 8. (a) Schematic diagram of Rh/Cr_2O_3 cocatalyst photodeposited on ($Ga_{1-x}Zn_x$)($N_{1-x}O_x$) photocatalyst, (b) Rh K edge XANES spectrum and (c) FT-EXAFS spectrum, (d) Cr K edge XANES and (e) FT-EXAFS spectrum for Rh/Cr_2O_3-($Ga_{1-x}Zn_x$) ($N_{1-x}O_x$). Reproduced with permission from Ref. [73].

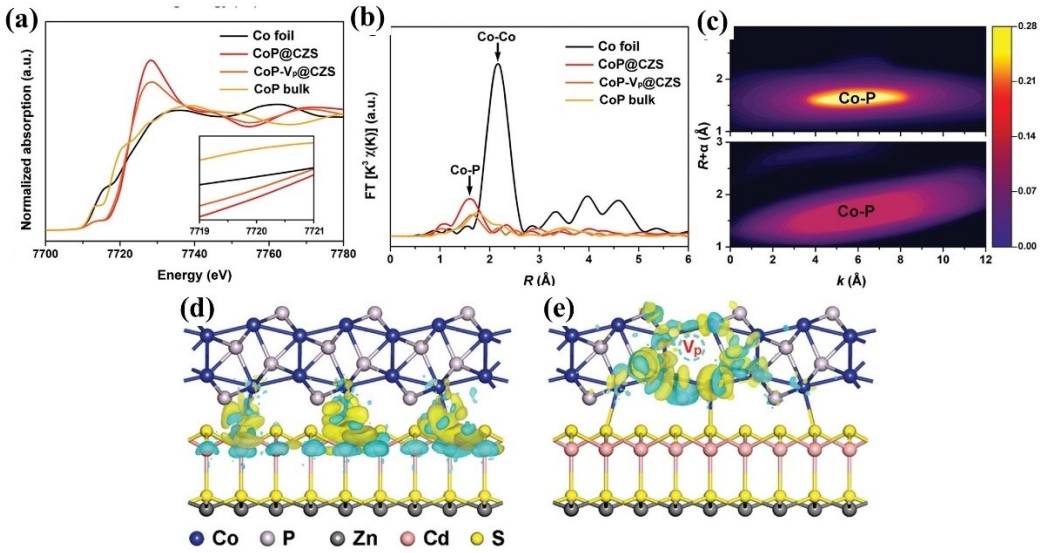


Figure 9. (a) Co K-edge XANES spectra, (b) FT-EXAFS spectra, (c) Wavelet transform plots, Schematic diagram of (d) CoP@$Cd_{0.5}Zn_{0.5}S$ and (e) CoP-V_p@$Cd_{0.5}Zn_{0.5}S$. Reproduced with permission from Ref. [74].

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the peak at 1.65 Å was corresponding to the Co–P coordination shells, meanwhile the Co–S bond was appeared and the intensity of Co–P bond was decrease in CoP–V_p@CZS compared with that in CoP@CZS, which indicated that CZS were anchored on CoP–Vp cocatalyst with single-atom phosphorus vacancies defects. As expected, the CoP–Vp cocatalyst with single-atom phosphorus vacancies defects could effectively accelerate the photogenreated charge carriers separation and transfer efficiency of CZS, which was certified by the DFT calculations (Figure 9d,e), the electrons tend to be aggregated in the CoP–Vp cocatalyst layer, which lead to the enhanced photocatalytic performance.[74]

The coordination environment and valence states of Pd in CdS@Pd-1.8 were clarified by XANES and EXAFS of the Pd K-edge in Figure 10. The XANES and EXAFS results disclose the huge difference between CdS@Pd-1.8 with the reference Pd foil and PdO, confirming the formation of Pd–S bond and PdS phase. The effective cocatalyst was PdS decorated on CdS, lead to accelerate the hole transfer process and boost the photocatalytic performance.[78]

3.1.2. Revealing the Location of Heteroatoms

The introduction of heteroatoms in the lattice of photocatalyst is one of the common and effective atomic-level regulative strategies, which cannot only accelerate the photogenerated charge carriers separation and transfer efficiency through the electronic interaction between heteroatoms and semiconductors, but also it can regulate the band gap of photocatalyst to improve the optical absorption through the introduction of impurity level.[79] Elemental specificity and atomic sensitivity make XAFS available to explore the local structure of heteroatoms, even in the ultra-low concentration of hetroaotms.[80] For instance, Zhao et al.[81] reveal the local coordination environments of B dopants in boron-doped g-C_3N_4 (CN–B) samples by XAS characterizations. As it descripted in Figure 11a, the triangular edge N (N1 site) and bridge N (N2 site) of g-C_3N_4 could be observed, which were showed at about 400.3 and 403.3 eV of the N K-edge XAS spectra, respectively. Meanwhile, the larger peak intensity ratio of N1/N2 indicated more defects at N2 sites. From C K-edge XAS spectra in Figure 11b, it can be observed that peak C_a was corresponding to the defect

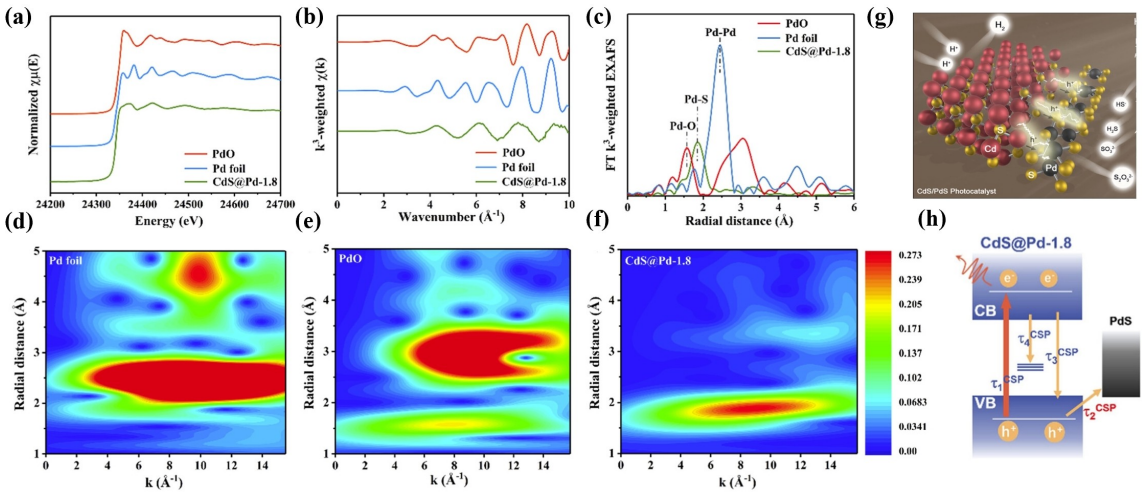


Figure 10. (a) Normalized XANES) spectra, (b) EXAFS spectra, and (c) FT- EXAFS spectra of Pd K-edge, (d–e) Pd K-edge Wavelet transform plots in CdS@Pd-1.8, Pd foil, and PdO, and (g–h) the photocatalytic mechanism of CdS@Pd-1.8. Reproduced with permission from Ref. [78].

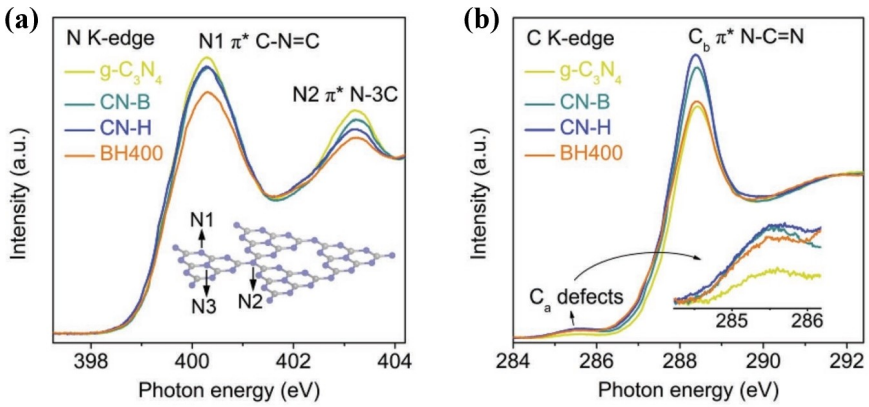


Figure 11. (a) N K-edge and (b) C K-edge XANES of g-C_3N_4, boron-doped g-C_3N_4 (CN–B), nitrogen-deficient g-C_3N_4 (CN–H), and BH400. Reproduced with permission from Ref. [81].

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structure in g-C_3N_4 framework. As presented in the inset of Figure 11b, peak C_a was corresponding to the defective structure in doped samples. While peak C_b was related to N–C=N sp_2 hybridized states. The declined intensity of peak C_b generally indicated the poor crystallinity or disordered atomic environment. The incorporation of B dopants creates abundant unsaturated sites and induces strong interlayer C–N interaction into g-C_3N_4, which effective tuned conduction and valence band positions, lead to the enhanced photocatalytic efficiency with 2.6 times.

Another example for the heteroatoms' incorporation of C_3N_4 is that Co doped into C_3N_4 nanosheets (CNNs) to synthesize Co/CN. The Co–N bonds formed in the Co–CNNs samples by the strong interaction of Co and CNNs contribute to improved photo-generated carrier separation and transport efficiency. The local coordination environment of Co in Co/CN were investigated by XAFS characterization. The results in Figure 12a–c, it can be observed that C atoms in g-C_3N_4 were replaced by Co atoms with appearance of two types of Co–N_x. One type is that each of the Co atoms is bonded with two in-plane N atoms; and the other type of Co–N_x is that each of the Co atoms is bonded with three N atoms of two neighboring CNN layers. The two types of Co–N_x caused both in-plane and interlayer charge transfer channels. Moreover, the Co doping also adjust the band structure and electronic structure of g-C_3N_4 to enhance light absorption and facilitate the separation and transfer efficiencies of photo-generated charges, lead to the improved photocatalytic hydrogen activity.[82] Tanaka and co-workers also used the XAS method to reveal the location of Rh dopant in Ga_2O_3 (Figure 12g–k). The XAS results found that after doping in Ga_2O_3 the trivalent Rh species substituted not for the tetrahedral Ga site but for the octahedral Ga site of β- Ga_2O_3, the d t_{2g} and e_g orbitals of octahedrally coordinated Rh^{3+} species caused two new dopant levels in the band gap of Ga_2O_3, further lead to the enhanced activity of photocatalytic CO_2 conversion.[83]

3.1.3. Elucidating the Interfacial Chemical Bond of Photocatalyst

It is well known that the tight interfacial contact, especially with the interfacial chemical bonding, is very essential to remarkably enhance the photocatalytic performance of composite photocatalyst by reducing the charge across resistance.[84] The interfacial electric field and interfacial chemical bonds between two composites are benefiting for accelerating the photo-generated electrons transfer through exploring and achieving precise electron-transfer channels to significantly improve the photocatalytic performance. However, the certification of interfacial chemical bonds in photocatalyst is challenging. Inhere, XAS has been extensively adopted to probe the fine structure of chemical bonding and unravel the chemical interfacial interaction in composite photocatalyst to clearly demonstrate their working mechanisms. For example, S K-edge and Pt L_3-edge XAFS characterizations were carried out to further reveal the local structure of CdS/PANI–Pt photocatalyst. As is shown in Fourier transformed EXAFS spectra (Figures 13a–c), it can be observed that the peak at about 2.33Å was corresponding to S–Pt bond, indicating a strong interfacial interaction existed between the CdS and Pt nanoparticles. The coordination number of Pt–S and Pt–N was plotted in the Figure 13d, it can be observed that the adequate and stable interfacial bonding with the suitable coordination number and coordination distance of Pt–S and Pt–N were found in CdS/PANI–O/Pt–OH

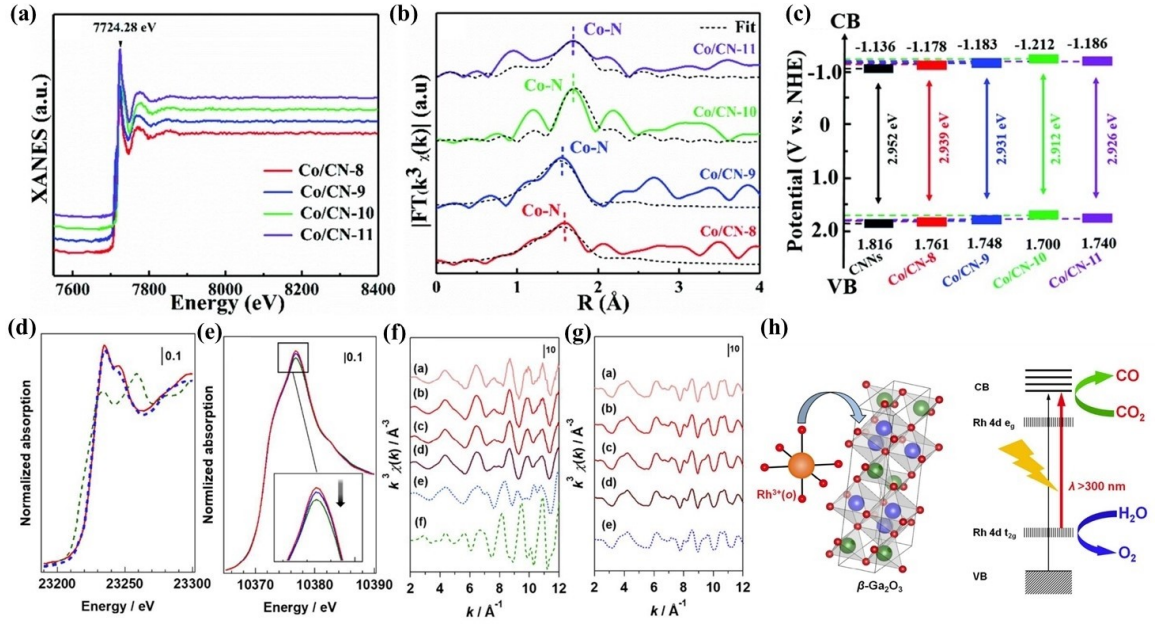


Figure 12. (a) XANES spectra, (b) EXAFS spectra, and (c) band structures of CNNs and Co/CN with different pH values. Reproduced with permission from Ref. [82]. (d) Rh K-edge XANES spectra of Ga_2O_3: Rh (red line) and reference samples, Rh_2O_3 (blue line) and Rh-foil (green line). (e) Ga K-edge XANES spectra of Ga_2O_3: Rh (n mol %) (n = 0.7 (red line), 3.2 (blue line), and 6.1 (green line)). (f) Rh K-edge and (g) Ga K-edge EXAFS of of Ga_2O_3: Rh, and (h) The band structure of Ga_2O_3: Rh. Reproduced with permission from Ref. [83].

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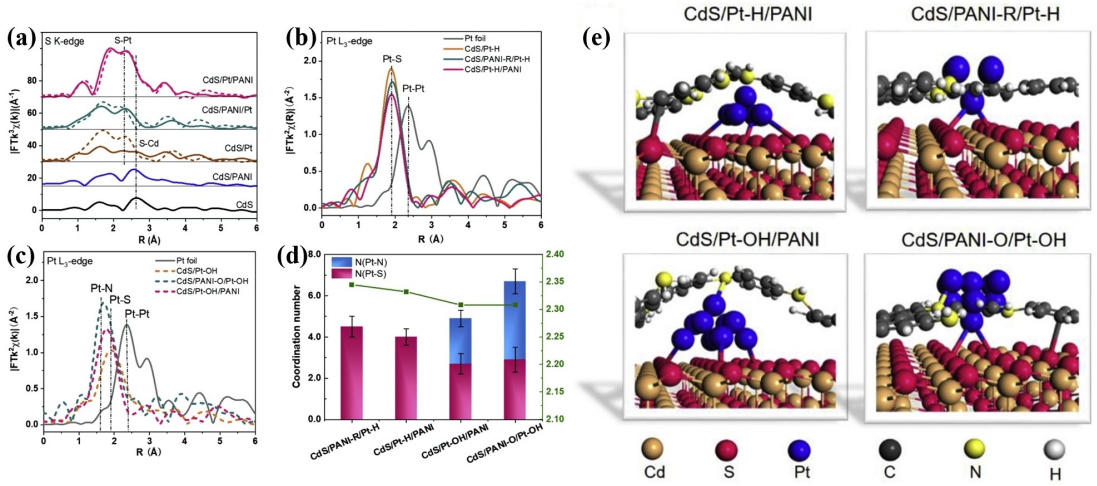


Figure 13. The EXAFS spectra in R-space of (a) S K-edge, Pt L_3-edge of PANI–Pt catalyzed CdS under (b) acidic and (c)alkaline condition, (d) The coordination numbers of Pt–S and Pt–N and the coordination distance of Pt–S, and (e) The demonstration of tunable interfacial structure in PANI–Pt catalyzed CdS systems. Reproduced with permission from Ref. [85].

sample, which was contributed to the improved photogenerated charge carriers' transportation by achieving precise electron-transfer channels with the formation of S–C, S–Pt, and Pt–N bond in the interfaces of CdS and the PANI–Pt cocatalyst, which enables the superior photocatalytic activity.[85]

A strategy to precisely construct the electron-transfer channels in g-C_3N_4/WO_3- bottom-up (CNWB) photocatalyst is to connect the valence band site of g-C_3N_4 with the conduction band site of WO_3 through the tungsten-nitrogen (W–N) bond. To explore accurately the local interface structure of CNWB, XAS was carried out. The W L_3-edge XANES spectra of CNWB and g-C_3N_4/WO_3-ultrasonic (CNWU) are shown in Figure 14a. The intensity of the absorption edge in CNWB was lower than that in CNWU, indicating that the oxidation state of W decreased due to the formation of N–W bonds. As shown in the FT-EXAFS spectrum from Figure 14b, the peak at about 1.70 Å was assigned to W–O bond in CNWU. Compared with CNWU, the peak at about 1.30 Å appeared in CNWB, which correspond to the W–N coordination, suggesting the formation of the W–N bond and the strong interfacial interaction between WO_3 and g-C_3N_4. The formation of W–N bond could act as atomic-level photogenerated electrons' channels to significantly accelerate photogenerated charge transfer, thus substantially boosting

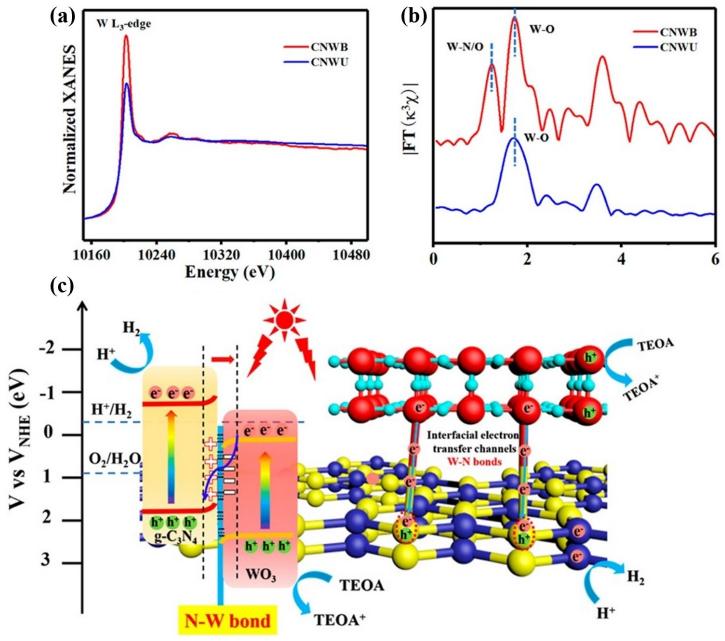


Figure 14. (a) XANES and (b) FT-EXAFS curves of g-C_3N_4/WO_3- bottom-up (CNWB) and g-C_3N_4/WO_3-ultrasonic (CNWU) at the W L_3-edge. (c) Photocatalytic reaction mechanism. Reproduced with permission from Ref. [86].

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photocatalytic efficiency, the photocatalytic enhanced mechanism with W–N bond was shown in Figure 14c.[86]

To accurately confirm the electronic structure and local coordination environment of Ta_3N_5/$NaTaO_3$ (N100-2 h), the Ta L-edge XANES spectra of N100-2 h were investigated, the results are shown in Figure 15a. The red shift in the absorption edge confirms the valence state of Ta in N100-2 h was increased compared with that in Ta foil. Meanwhile, the EXAFS spectra is depicted in Figure 15b. A new peak at about 1.7 Å appeared, which was ascribed to the formation of O–Ta–N bond. In addition, the fitting of Ta L-edge EXAFS spectra of N100-2 h (Figure 15c–e) further confirm the formation of O–Ta–N bond. Figure 15f schematically illustrates the O–Ta–N connections involved in the interfacial structure, which significantly accelerate the photogenerated charge transfer, leading to the excellent photocatalytic performance. The H_2 evolution rate and O_2 evolution rate for the N100-2 h sample are 25.12μmol/h and 12.48 μmol/h, respectively, which are 31.62 times higher than those for $NaTaO_3$.[87]

To further validate the precise interfacial local coordination environment of Co and Fe atoms in photocatalyst MIL-125-NH_2@CoFe PBA, XAS characterization was carried out and the results shown in Figure 16. As shown from the Ti K-edge EXAFS spectrum in Figure 16a, the peak at about 3.6 Å could correspond to the Ti–O–Co coordination. As is depicted from Co K-edge EXAFS spectrum in Figure 16b, the peak at about 3.6 Å also assigned to the Ti–O–Co coordination. Furthermore, wavelet transform (Figure16c–f) was further analyzed to clarify the local structure of the MIL-125-NH_2@CoFe PBA heterojunction with the formation of Ti–O–Co bond. The interfacial built-in electric field caused by the formation of Ti–O–Co bonds could effectively accelerate photogenerated charge carriers'

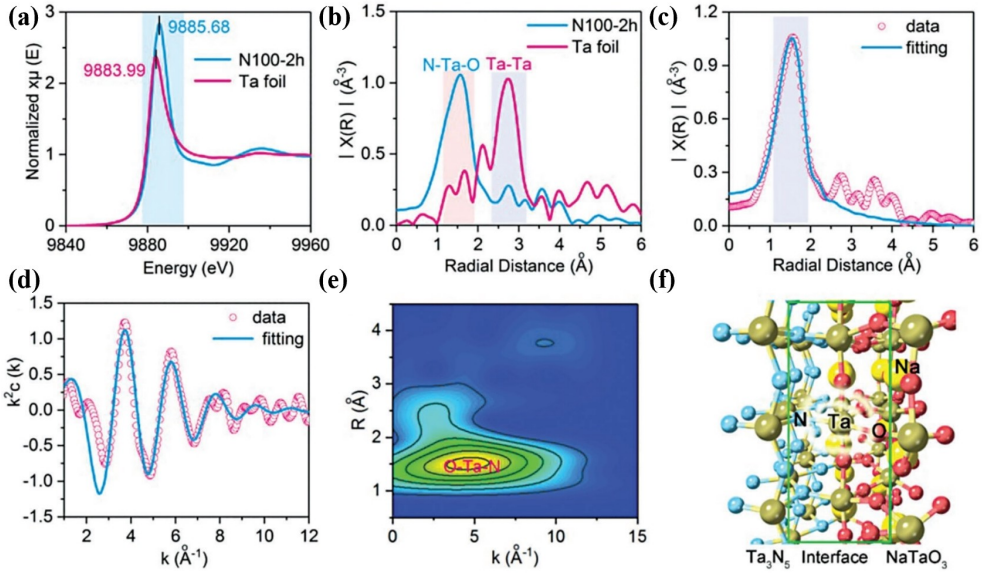


Figure 15. (a) Ta L-edge XANES spectra and (b) the corresponding Ta L_3-edge EXAFS spectra of Ta foil and N100-2 h, the fitting of Ta L-edge EXAFS data for the N100-2 h sample in (c) R space and (d) k space, (e) The wavelet transform of N100-2 h, and(f) Schematic illustration of the crystal structure of the N100-2 h heterojunction with the interface connected by O–Ta–N. Reproduced with permission from Ref. [87].

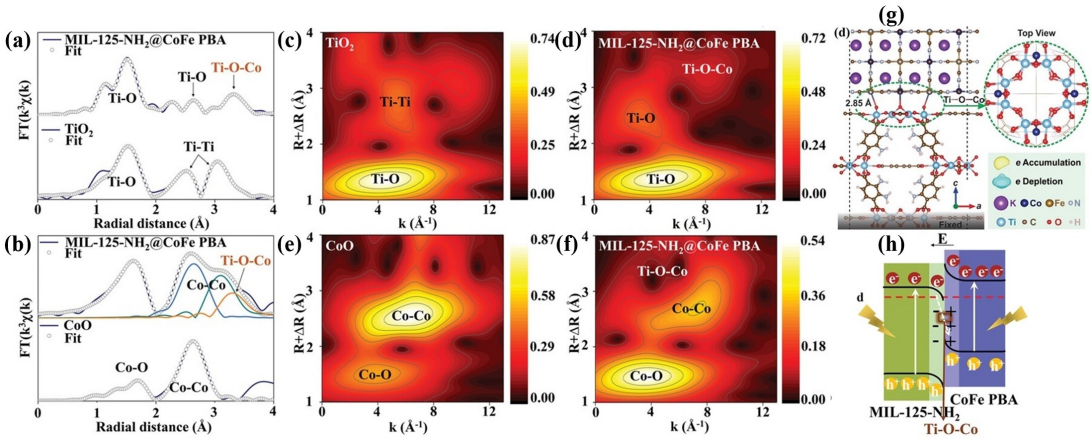


Figure 16. EXAFS spectra for (a) Ti K-edge and (b)Co K-edge of MIL-125-NH_2@CoFe PBA, (c, d) Ti K-edge wavelet transforms plots and (e, f) Co K-edge wavelet transforms plots, (g) Optimized structure of MIL-125-NH_2@CoFe PBA heterojunction, (h) Schematic illustration of charges transfer mechanism of MIL-125-NH_2@CoFe PBA. Reproduced with permission from Ref. [88].

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transportation, the excellent photocatalytic efficiency is increased by 3.2 times.[88]

A heteroatom-induced interface construction strategy was carried out to fabricate nanosheet heterojunctions with titanium-organic frameworks (MOF-902) and thiophene-based covalent triazine frameworks (CTF–Th). The S atom in thiophene could bonded with Ti atoms to fabricate the T–S bond. The interfacial Ti–S bond could significantly shorten the charge carrier migration pathway from MOF-902 to CTF–Th under illumination. As shown in the Figure 17a, the *K*-edge XANES spectra reveal the oxidation state of Ti and the gradual rise of the white line intensity upon composite formation. The direct evidence of the coordination structure of the MOF-902@CTF-$Th_{1.5}$ can be observed from the EXAFS spectra in Figure 17b. The peak at about 1.5 Å was ascribed to the Ti–O first coordination shell, another peak at 2.1 Å was associated with Ti–S bond. The existence of Ti–S bond in the interface of MOF-902@CTF-$Th_{1.5}$, further facilitated the charge transfer of S-scheme heterojunction, the photocatalytic mechanism was shown in Figure 17c.[89]

3.1.4. Identifying the Phase Transformations

Photocatalysts often undergo structural and phase transformations when they are in the preparation process. These transformations are associated with their evolving photocatalytic performance, and a fundamental understanding of the photocatalytic process is vital for the rational design of photocatalyst. Inhere, the phase transformation of photocatalyst can be probed by XAS method. For instance, the Zn K-edge XANES spectrum is applied to investigate to the phase transformation from ZnS to ZnO in Figure 18. As shown in Figure 18a, the peak at about 9664 eV was ascribed to sphalerite ZnS. And when the temperature increased up to 500 °C, a new peak at 9677 eV was appeared, which was explained by the incipient formation of ZnO, indicating that the oxidation of ZnS has begun at 500 °C. At 550 °C and 600 °C, the obtained XANES spectra are comparable with that of ZnO, confirming that the complete transformation of ZnS to ZnO. The phase transformation of photocatalysts or even the creation of a ZnS/ZnO heterojunction significantly increased the photoexcited charge carriers' transfer and photocatalytic performance of photocatalyst.[90]

3.1.5. Uncovering the Local Structures of Single Atomic Photocatalysts

Due to the maximum atomic utilization efficiency and distinct advantages of outstanding catalytic activity, single-atom photocatalysts have emerged as promising candidates for photocatalysis. The introduction of isolated metal atoms could modulate the band structure to enhance the light-harvesting capability of semiconductor and also serve as electron/holes traps to boost the transfer of photoexcited charge carriers, thus significantly enhanced the photocatalytic performance of the single-atom photocatalysts.[91] Furthermore, the structural simplicity of single-atom photocatalyst allows us to explore precise structure-activity relationships at the atomic scale, and to investigate deeply the fundamental mechanism of photocatalysis. Therefore, identifying the local structure of isolated metal atoms in single-atomic photocatalyst is essential to understand their phtocatalytic reaction mechanism. However, traditional characterization methods in analyzing single-atomic photocatalyst face huge challenges, such as the much less atomic concentrations and smallest size. In this regard, XAS can be obtained in one test to analyze both electronic and coordinate structures of single atomic photocatalyst. g-C_3N_4 is an excellent support for single-metal atoms because of its ultrahigh content of lone pairs of electrons and good coordination ability with isolated metal atoms. Meanwhile, the metal-nitrogen (M–N) coordination configuration could effectively facilitate the photogenerated charge carriers transfer. For instance, the valence state and coordination environment of Ni on g-C_3N_4 are examined by XANES spectroscopy and EXAFS spectroscopy. As shown in Figure 19a, the pre-edge absorption of Ni coordinated pyridinic nitrogen/g-C_3N_4 (Ni–N/CN) was close to that of NiO and higher than that of Ni foil, suggesting that the oxidation state of partial Ni is +2. The EXAFS spectrum of Ni–N/CN is shown in Figure 19b, the peak at 1.49 Å was assigned to the Ni–N interaction in the first coordinating sphere. The absence of Ni–Ni peaks further confirmed the mononuclear nature of

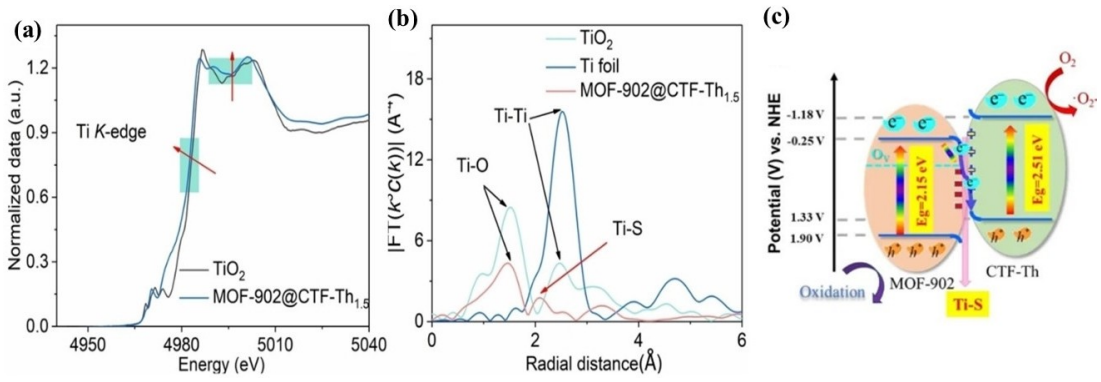


Figure 17. (a) XANES spectra, (b) The k^3-weighted Fourier transform spectra derived from EXAFS spectrum of TiO_2, Ti foil and MOF-902@CTF-$Th_{1.5}$, and (c) photocatalytic mechanism for MOF-902@CTF-$Th_{1.5}$. Reproduced with permission from Ref. [89].

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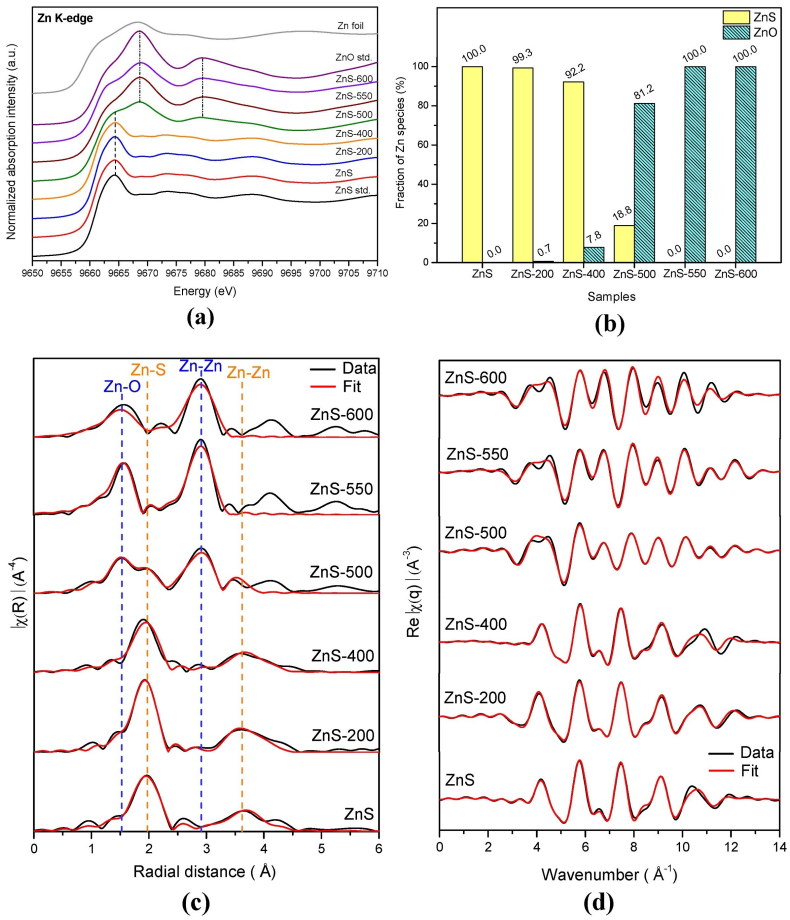


Figure 18. (a) The normalized absorption Zn K-edge XANES spectra, (b) the LCF results of the fraction of Zn species (%), the experimental EXAFS fitting spectra in R-space (c) and q-space (d). Reproduced with permission from Ref. [90].

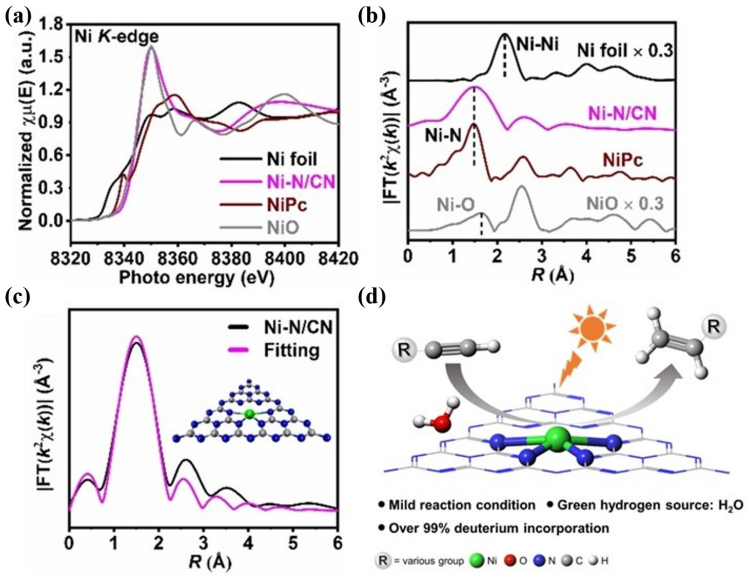


Figure 19. (a) XANES spectrum at the Ni K-edge, and (b) Fourier-transform EXAFS spectrum at the Ni K-edge with reference to Ni foil, NiO, and NiPc. (c) Fitting of the EXAFS spectrum at the R space. Inset: illustration of the Ni–N4 configuration (Ni (green), N (blue), C (gray)); (d) The proposed visible-light-driven semihydrogenation mechanism mediated by single-atom Ni. Reproduced with permission from Ref. [92].

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the single-atom Ni site, specifically, the four-coordinated and isolated Ni^{2+}–N_4 site on the CN scaffold. The Ni^{2+}–N_4 site paired up with pyridinic N of a CN scaffold, is illustrated in Figure 19c (inset), which enabled hydrogen delivery from water to alkynes, ensuring excellent semihydrogenation selectivity.[92]

The Fe K-edge XAS characterization was carried out to further reveal the local structure of Fe in Fe_1-g-C_3N_4 in Figure 20, it can be observed from Figure 20a that the white line peak of Fe_1-g-C_3N_4 located between those of reference FeO and Fe_2O_3, indicating the valence state of Fe was between +2 and +3. From the FT-EXAFS of Fe-g-C_3N_4 in Figure 20b, the significant shift from the nominal 2.28–2.51Å could be ascribed to Fe embedding into the center of the six-fold cavity of g-C_3N_4. Quantitative EXAFS curve-fitting analysis (Figure 20c) determines the single-atomic Fe atoms are most probably connected to two unsaturated N atoms with the typical M–N_2 motif in Fe_1-g-C_3N_4. In Figure 20d, it can clearly observe that the isolated Fe–N_2 coordiantion was connected to the edge of cavity structure of g-C_3N_4 framework in the Fe_1-g-C_3N_4 sample. The targeted Fe_1-g-C_3N_4 exhibited the excellent photogenerated electron-hole separation and transportation efficacy, lead to the improved photocatalytic H_2 production performance.[93]

Anchoring the single-atom Nb to g-C_3N_4 resulted in the existence of N defects, which led to the formation of Nb–C coordination bond, further leading to the faster photogenertaed charge carriers transfer. The electronic states and coordination environments of the Nb species in g-C_3N_4/Nb0.06 (CNNb0.06) are further examined using synchrotron radiation-based XANES and EXAFS characterizations. From the XANES spectrum of Nb K-edge in Figure 21a, the white line peak of CNNb0.06 was found between Nb foil and Nb_2O_5, indicating that the valence state of Nb in CNNb0.06 ranged from 0 to +5, with the predominance valence state of +5. From the EXAFS of Nb K-edge in Figure 21b, the formation of Nb–C bond could be speculated due to the single-atomic Nb was replaced with N. Besides, as displayed in Figure 21c, quantitative EXAFS curve fitting indicated that the single Nb atoms are replaced with bridged N and coordinated with three carbon atoms. The wavelet transform in Figures 21d–f further certified the formation of Nb–C bond, suggesting that the single-atomic Nb was connected with C atoms in the framework of g-C_3N_4 support for CNNb0.06, leading to the faster photogenertaed charge carriers transfer and the excellent photocatalytic performance.[94]

XAS analysis was carried out to reveal the local structure of Cu atoms in SA–Cu–CN-620 sample. From the XANES spectrum of Cu K-edge (Figure 22a), it can be observed that the valence state of Cu in SA–Cu–CN-620 was between 0 and +1. Then the local structure of the Cu atom was further investigated by FT-EXAFS, the results are shown in Figure 22b, the peak at 1.49 Å was ascribed to the Cu–N/C bond, moreover, no Cu–Cu bond and Cu–O–Cu bond arose in SA–Cu–CN-620. These results indicated that the Cu atoms are highly dispersed in the single-atomic form without Cu or CuO phase, further confirmed by the EXAFS wavelet transforms from Figure 22c. Moreover, the EXAFS fitting provided the quantitative local structure information in Figure 22d, the coordination structure of Cu is Cu–N_3C_1, where Cu is simultaneously coordinated with three unsaturated-coordinated N atoms and one saturated-coordinated C. The specific coordination structure of Cu–N_3C_1 is displayed in the inset of Figure 22e, it can be observed that the single-atomic Cu was bridged between C_3N_4 interlayers. These local coordination structure is benefit to the photogenerated charge carriers' transportation, lead to the improved photocatalytic hydrogen evolution performance.[95] Similarly, Zou and co-

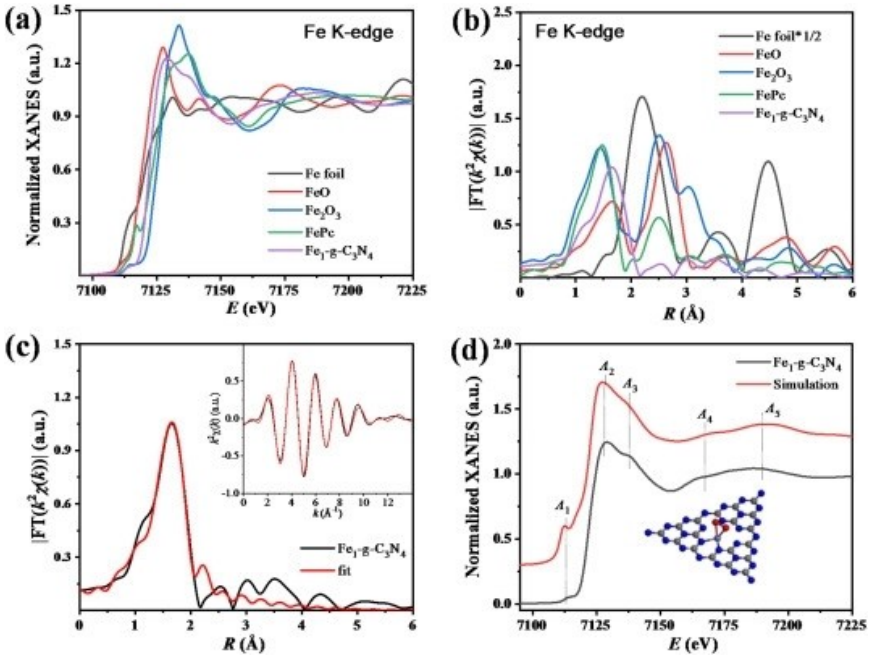


Figure 20. (a) Fe K-edge XANES spectra, (b) FT-EXAFS spectra in the R-space, and (c) EXAFS curve-fitting analysis in R-space of Fe_1-g-C_3N_4. (d) Comparison between the XANES experimental spectrum of Fe_1-g-C_3N_4 and the theoretical spectrum calculated based on the Fe_1–N_2O_2 coordination (inset structure). Reproduced with permission from Ref. [93].

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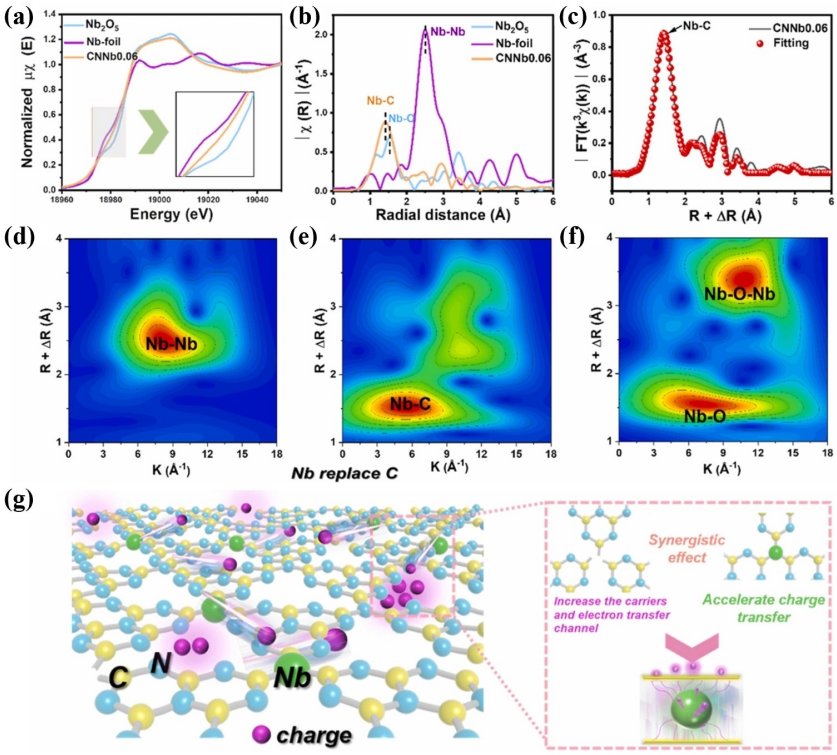


Figure 21. (a)Normalized Nb K-edge XANES spectra, (b)FT-EXAFS spectra, (c)FT-EXAFS fitting curves in R space of g-C_3N_4/Nb-0.06 (CNNb0.06); (d–f) the WT-EXAFS plots of Nb-foil, CNNb0.06, and Nb_2O_5, respectively; (g) the illustration of local structure of CNNb0.06. Reproduced with permission from Ref. [94].

workers utilized the XANES and EXAFS spectroscopy to analysis the coordination structure on single indium atom decorated carbon nitride (ICN) photocatalysts. As it depicted from the FT-EXAFS spectrum in Figure 22f, the peak at about 1.6 Å was ascribed to the In–N coordination, indicating that isolated In atoms are bonded with the unsaturated N atoms in g-C_3N_4. The coordination number of In–N bond was about 4, suggesting that the single In atom was coordinated by four N atoms in the matrix of g-C_3N_4, the corresponding $In^{\delta+}$–N_4 configuraion reduces the (002) interplanar spacing of *g*-C_3N_4 by electrostatic adsorption. As displayed in Figure 22g, the $In^{\delta+}$–N_4 sites could not only facilitate the photogenerated electrons transportation, but also served as the active sites to adsorb and activate CO_2 during CO_2 photoreduction, which significantly promotes the reactivity of CO_2 photoreduction.[96]

3.2. Unraveling the in situ Structure and Chemical State Evolution of Photocatalysts

Probing the dynamic evolution of catalyst structure and chemical state under operating conditions is highly important for investigating the reaction mechanism of catalysis more in depth, which in turn advances the rational design of redox catalysis in using renewable energy to produce fuels. In addition to ex-situ XAFS characterization, in-situ XAFS characterization has also been carried out to study the electrochemical reactions and dynamics of materials in real time. XAFS conducted in situ or operando is a valuable tool to probe the dynamic electronic and coordination structure of photocatalyst under working conditions, but care must be taken when designing the experiments. To better understand the structure-performance relationship of photocatalyst, the in-situ XAS technologies were utilized to track the changes of catalyst structure and chemical state of photocatalyst in the working state. For example, the Cu and Ni K-edge XANES spectra are analyzed to reveal the chemical state evolution of Cu and Ni in 5Ni5Cu–TiO_2 photocatalyst during the photocatalytic process. As shown in Figure 23a and b, the absorption edge of Ni K-edge at lower energy was lower, and the absorption edge of Cu K-edge at lower energy was higher in the 5Ni5Cu–TiO_2 sample, indicating that the 5Ni5Cu sample contains a larger proportion of metallic Cu and a smaller proportion of metallic Ni. Subsequently, in situ XAS experiments are conducted on 5Ni5Cu–TiO_2, the Ni and Cu K-edges XANES spectra are acquired at different time intervals under dark conditions (Figure 23c–d) and under illumination (Figures 23e–f), which enables real-time monitoring of chemical changes occurring in the 5Ni5Cu co-catalyst. The in-situ XANES results as evident from Figure 23 g–h indicate that: (i) Ni and Cu are partially oxidized in the liquid phase under dark conditions in the 5Ni5Cu co-catalyst; (ii) the formed Cu oxides and Ni oxides dissolved in liquid of ethanol-water; (iii) a UV light photo-induced redeposition on the TiO_2 surface of both dissolved Cu^{2+} and Ni^{2+} from the solution under illumination. The XAS results not only prove a photoinduced regeneration of the NiCu co-catalyst but also explicit that the phases of the metallic states of Ni and Cu are the active sites and responsible for the

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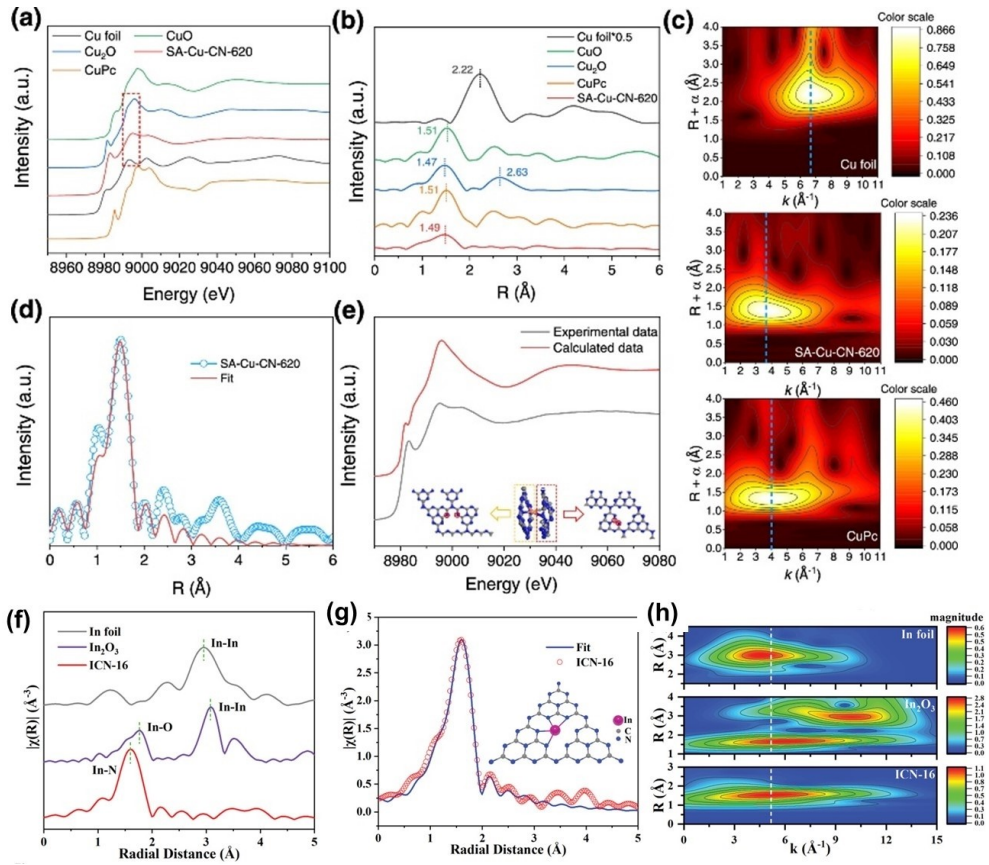


Figure 22. (a) Cu K-edge XANES spectra and (b) FT-EXAFS in R space for SA–Cu–CN-620 and control samples. (c) WT-EXAFS plots. (d) EXAFS fitting curve at R space for SA–Cu–CN-620. (e) Experimental (black) and calculated (red) Cu K-edge XANES spectra of SA–Cu–CN-620. Reproduced with permission from Ref. [95]. (f) The In K-edge FT-EXAFS spectra, (g) EXAFS fitting for single indium atom decorated carbon nitride (ICN-16), the inset is structure model of $In^{\delta+}$–N_4 site. (h) WT-EXAFS plots. Reproduced with permission from Ref. [96].

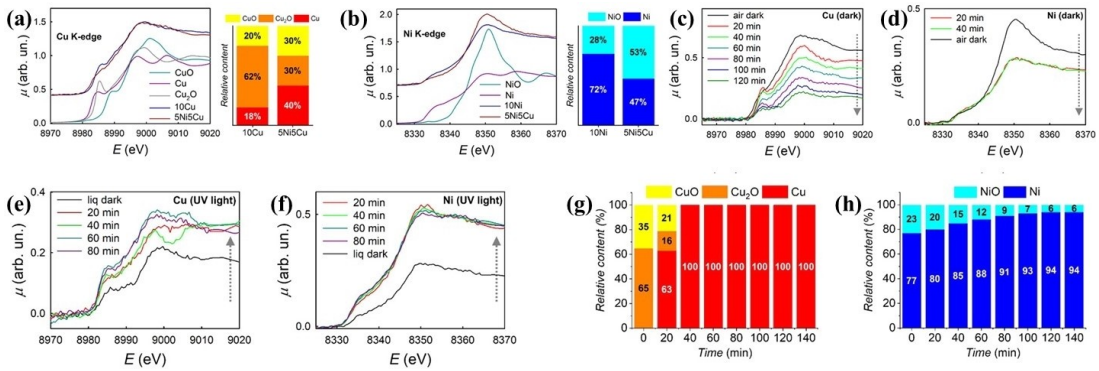


Figure 23. (a) Normalized Cu K-edge XAS spectra and (b) Normalized Ni K-edge XAS spectra exposed to air in the dark. (c) In situ Cu K-edge and (d) In situ Ni K-edge XAS spectra of 5Ni5Cu–TiO_2 in the dark for different times. (e) In situ Cu K-edge and (f) In situ Ni K-edge XAS spectra of 5Ni5Cu–TiO_2 under UV light illumination for different exposure times; (g) Cu and (h) Ni phase compositions calculated from (c–f). Reproduced with permission from Ref. [70].

significantly improved photocatalytic H_2 evolution of NiCu cocatalyst in 5Ni5Cu–TiO_2 photocatalyst.[70]

Another interesting ex-situ and in-situ XANES study allowed to characterize valence state and structural evolution of Cu species, to further establish the realistic structure-performance relationship and photocatalytic mechanism of Cu-based photocatalyst. First, from the ex-situ Cu K-edge XANES spectra in Figure 24a–b, it can be observed that the valence states of Cu in photocatalyst Cu doped in TiO_2 (Cu-mT) are mainly +2 due to the 1 s→3d feature of Cu^{2+} species at 8977 eV. It's worth noting that the peak at 8983 eV of Cu K-edge XANES in 1Cu-mT, which was ascribed to 1 s→4p transition, exhibited a red shift compared with that in $Cu(OH)_2$, suggesting that partial Cu^{2+} are in a distorted site, which was also confirmed by the FT-EXAFS spectra (Figure 24c). To in-depth understanding the structure-performance relationship, in addition to ex situ XAFS character-

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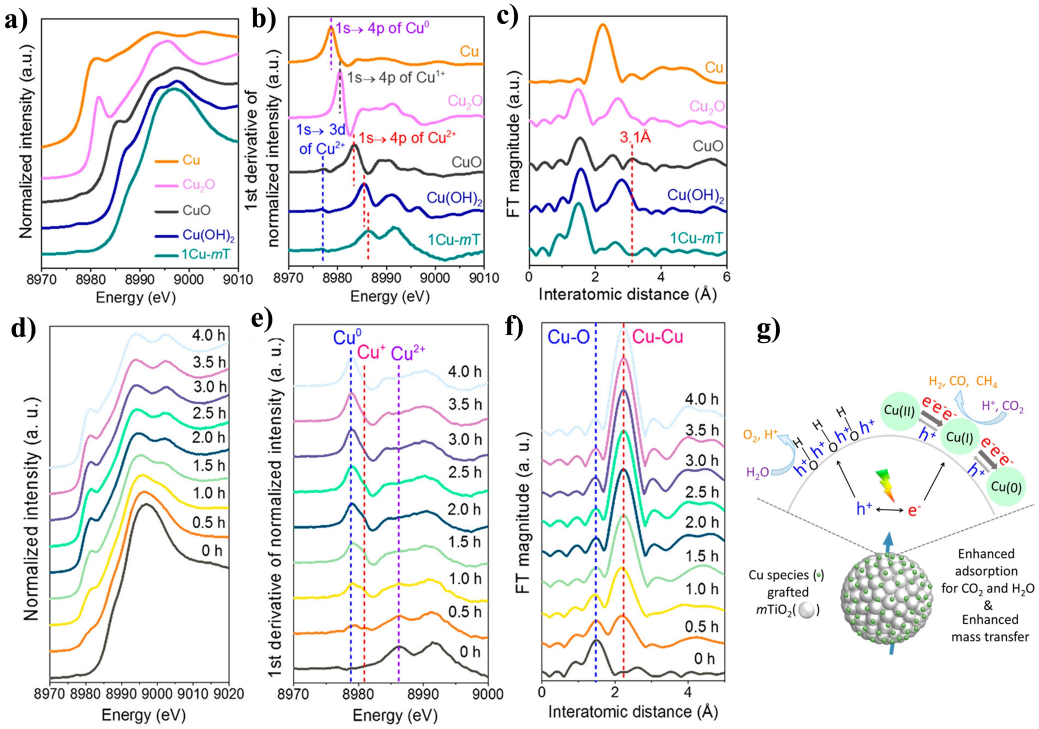


Figure 24. (a) Cu K-edge XANES spectra, (b) the first derivatives from (a), and FT-EXAFS spectra for reference materials and Cu doped in TiO_2 (Cu-mT). (d) In situ Cu K-edge XANES spectra, (e) The first-derivative from (d) and (f) EXAFS spectra for 1Cu-*m*T under photocatalytic process. (g) Probable reaction mechanisms for photocatalytic reduction of CO_2 with H_2O over the Cu-*m*T catalysts. Reproduced with permission from Ref. [97].

ization, in situ XAFS characterization has also been carried out to study the valence state and structural evolution of Cu species of 1Cu-mT in real time during the photocatalytic CO_2 reduction (Figure 24d–e). It shows that the valence state of Cu species in fresh 1Cu-*m*T sample is +2 before light illumination, then light illumination induces several modifications in the spectra: the peaks for metallic Cu^0 and Cu^+ are appeared after illumination for 0.5 h, while that for Cu^{2+} significantly decreased, indicating light promotes the reduction of the Cu^{2+} to Cu^+ or Cu^0, which take placed by the photogenerated electrons in the he CB of TiO_2. The in-situ EXAFS in Figure 24f further confirms the gradual transformation of Cu^{2+} to Cu^0. These mixed phases of Cu^+/Cu^0 are the active centers for photocatalytic CO_2 reduction. Moreover, the reductions of $Cu^+/Cu^0$and Cu^{2+}/Cu^+ both are reversible. It can be observed that Cu^0 could be oxidized to Cu^+ with the photogenerated holes. The reversible transformation of $Cu^{2+}/Cu^+/Cu^0$ could significantly accelerate the photogenerated charge carriers' separation and transfer. The possible reaction mechanisms of the Cu-mT for photocatalytic CO_2 reduction could be schematically illustrated in Figure 24g. Upon light illumination, the electron was excited from the VB to CB of $mTiO_2$, the initial Cu^{2+} species can be gradually reduced to Cu^+ by the electrons and further to Cu^0, the mixed states of Cu^+/Cu^0 mixture could not also significantly improve the CO_2 adsorption ability, but also accelerate photogenerated charge carriers transportation, both leading to the enhanced photocatalytic CO_2 reduction activity.[97]

Another example of phase transformation is that from Liu et al. identify the local structures and the phase transformations of the photocatalyst using synchrotron radiation XANES and EXAFS tests at the Ru K-edge. Based on the XPS result in Figure 25a, as it displayed from the XANES spectrum in Figure 25b, it can be observed that the red shift in the absorption edge within the increased oxidation time, indicating that the oxidation states of Ru are increased due to the insertion of oxygen atoms. The corresponding FT-EXAFS spectra are shown in Figure 25c, the two dominant peaks at about 1.56 Å and 2.40 Å of Ru–RuO_x NPs/C_3N_4 (RRCN) sample are ascribed to the Ru–O and Ru–Ru coordination, indicating the coexistence of Ru and RuOx phases. Combined with the XANES results, it can be predicted that the oxygen is inserted into the Ru nanoparticles to form RuO_x in RRCN samples. Furthermore, the local structural evolution of the Ru–RuO_x was further certified by EXAFS in Figure 25d. Therefore, the amount of RuO_x component in the Ru–RuO_x heterostructures can be well tuned by controlling the heating time, then Ru–RuO_x heterostructures could lead to the high photocatalytic performance.[98]

3.3. Ascertaining the Charge Rransfer Process of Photocatalysts

Although carrier behavior in photocatalysts has been discussed in terms of the band theory and their energy levels in reciprocal space (k-space) based on optical spectroscopic results, it has rarely been discussed where photocarriers are located in real-space (r-space) based on direct observation of the excited states.[99] XANES can be used to evaluate elementary electronic

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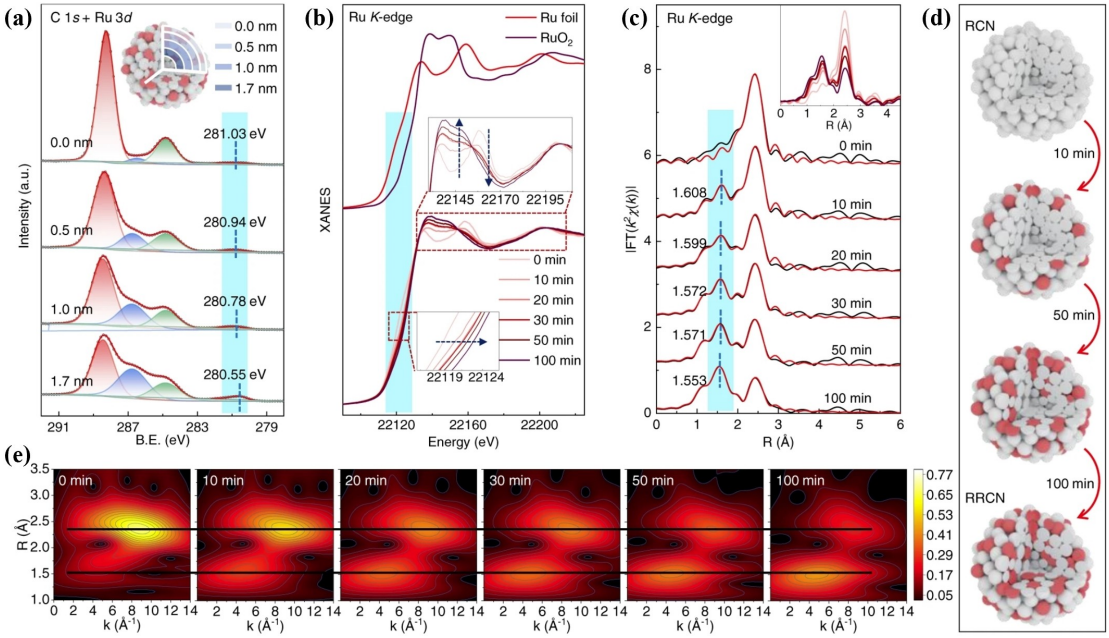


Figure 25. (a) (C 1s + Ru 3d) XPS of the as-synthesized samples (b) In-situ XANES spectra recorded at the Ru *K*-edge of Ru–RuOx NPs/C_3N_4 (RRCN) at different oxidation time during the heat process. (c) in-situ EXAFS fitting spectra at the Ru *K*-edge. (d) Schematic illustration of the oxygen insertion process during the formation of RRCN. (e)Wavelet analysis of the RRCN sample during in-situ EXAFS states. Reproduced with permission from Ref. [98].

structures directly and charge transfer indirectly.[100] Time-resolved XAFS measurements in a wide range time domain and with varied wavelengths of the excitation pump laser facilitate understanding the local electronic and geometrical structures evolution around the X-ray absorbing atom and can address photocarrier dynamics that have a significant influence on the photocatalytic performance.[101]

Synchrotron XAS experiments were also performed to ascertain electronic transfer processes during CO_2 photocatalytic reduction over Ag/TiO_2 and Au–Ag/TiO_2. The Ti K-edge XANES spectra of Ag/TiO_2 were acquired under UV-irradiation (Figure 26a), and although in general terms, the profile is rather similar to that obtained in the dark, a clear decrease of the intensity of the pre-edge features is observed, which indicates lower density of unoccupied states ascribed to an increased electron density in the d-states. Therefore, the results prove a charge transfer directly into d (t_{2g} and e_g) states after illumination of the solid with UV light, which can be corresponded to the photoinduced transference of electrons from the valence band to the conduction band of TiO_2. In parallel, more conspicuous changes were observed in Ag K-edge spectra during cycles of on/off UV external irradiation under reaction conditions (a flow of CO_2 and H_2O vapor), as displayed in Figure 26b. Once stabilized under the X-ray beam, Ag is present as a mixture of Ag^0/Ag^+, but under UV light, the edge experiences a red shift, which is related to an additional reduction of Ag. In addition, the intensity of both the white line and the continuum resonances varies reversibly with irradiation. These observations suggest that a photoinduced charge transfer from TiO_2 to Ag states takes place under UV irradiation, but

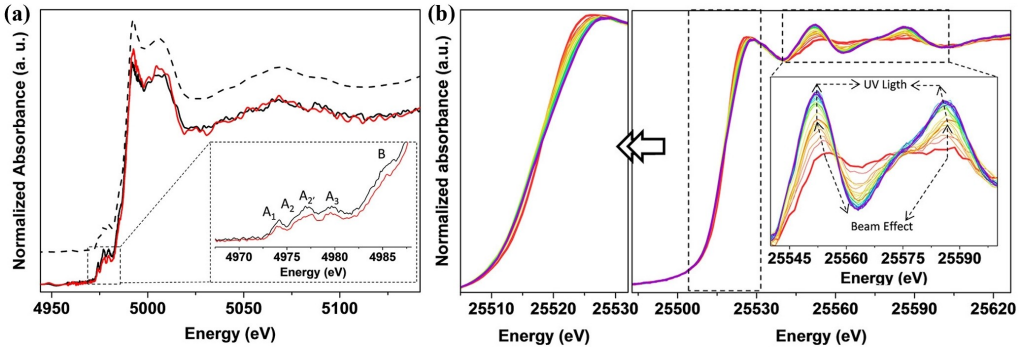


Figure 26. (a) Ti K-edge XANES spectra of TiO_2 (dashed trace), as well as Ag/TiO_2 before (black trace) and after (red trace) UV illumination under reaction conditions. (The pre-edge structure of Ti cations in octahedral sites presents four peaks: A_1, A_2, A_3, and B. A_1 is mainly attributed to quadrupolar transitions to t_{2g} states of the TiO_6 octahedron. The second and the third features, A_2 and A_3, are attributed to 1 s–3d dipolar transitions (to t_{2g} and e_g states) and the shoulder B is assigned to the 1 s-to-4p transition.); and (b) Ag K-edge XANES spectra and magnification (left and inset) during exposure to the X-ray beam and during UV light switching under reaction conditions. Reproduced with permission from Ref. [102].

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this process is fully reverted in the dark. The electrons accumulated in the Ag-sites (electronically coupled to the Ti atoms of the titania support) are used for photocatalytic CO_2 reduction, favoring the formation of highly reduced products such as methane.[102]

Our study utilized XAS and in-situ XAS techniques to investigate the electronic and local atomic structures of Au THMS samples with varying Au concentrations under visible and UV-visible illumination. The XAS results at Ti L- and K-edges revealed a strong influence of Au nanoparticle size on the symmetry of TiO_6 octahedra and the unoccupied state of Ti_{3d} t_{2g} state (Figure 27). The results indicated that the size of Au NPs varied with different Au concentrations. Moreover, it significantly affected the electronic structure and showed a strong correlation with the local structural symmetry. In-situ XAS results also demonstrated changes in the electronic structure and charge transfer under visible and UV-visible illumination, particularly in the case of the Au THMS (0.5 % Au) sample, which enhanced its photocatalytic activity. The in-situ XAS at Au L3- and Ti L3-edges confirmed the hot electron charge transfer from Au 5d to Ti 3d t2 g state due to the LSPR effect under visible-light illumination. A comparison of the XAS and PEC performance indicated that lattice distortion and corresponding symmetry changes, along with the size of Au nanoparticles, significantly influenced the rate of hot electron charge transfer, thereby resulting in variations in PEC activity of AuTHMS samples depending on the Au content, with higher activity observed in the case of the 0.5 % Au sample.[103]

3.4. Revealing the Photoexcitation Processes of Photocatalysts

The discussion of photogenerated charge carriers' behavior in photocatalyst has primarily been approached from a band theory perspective.[104] During the photoexcited process, whether electrons in the conduction band (CB) or holes in the valence band (VB), transfer to the surface of the photocatalyst and react with substances.[105] Inhere, XAFS can be used to study photoexcitation dynamics in real space. Understanding the photoexcitation mechanism of photocatalyst is crucial for the development of more efficient photocatalysts for visible light utilization. It is necessary to elucidate the local structure and electronic state of the photoexcited state in order to comprehend the photocatalysis process and enhance the photocatalytic performance.

In regard to the energy band structure of $BiVO_4$, Bi atoms is crucial to monitor the excitation and transition of photogenerated electron, however, the local structure evolution of Bi in the photoexcitation process of $BiVO_4$ has not been systematically explored. In terms of this issue, XAFS characterization can be employed to monitor the local and electric structural changes of Bi atoms in $BiVO_4$ after photoexcitation to reveal the local electronic structure of $BiVO_4$. Recently, Uemura et al. have investigated the photoexcited state of $BiVO_4$ using Bi L_3-edge XANES spectroscopy (Figure 28). Figure 28a depicts the Bi L_3 edge XANES of $BiVO_4$ in its ground state, as well as the difference spectra after photoexcitation. Upon photoexcitation, the difference of peak B in the Bi L_3 edge difference XAFS spectrum could be ascribed to the dynamic local structural change around Bi atoms. The appearance of peaks A and C could be attributed to a static distortion in the Bi structure, as the Bi–O bond distance primarily affects the Bi L_3 edge feature. After several tens of picoseconds, the photoexcited electrons at V 3d were stabilized by the metastable state of $BiVO_4$. The photoexcitation process of $BiVO_4$ could be predicted from these differences in the Bi L_3-edge XANES spectra after photoexciation, which was presented in Figure 28b. After light irradiation on $BiVO_4$, photogenerated electrons were excited from the VB (O 2p) to the CB (V 3d). Simultaneously, the dynamic fluctuation of Bi atoms caused by the coherent oscillation between Bi^{3+} and VO_4 occurs below 2 ps. Subsequently, the initial excited state decayed to a metastable state with a distortion structure in Bi–VO_4 units with a short time (about 14 ps). Moreover, the partial distortion could be stabilized with the photogenerated holes with about 40 ns.[106]

In addition to $BiVO_4$, Uemura et al. also reported the photoinduced processes of WO_3 through pump-probe XAFS technique by study the photoexcitation states of WO_3.[107] The O 2p orbitals contributed to the valence band of WO_3, while W 5d

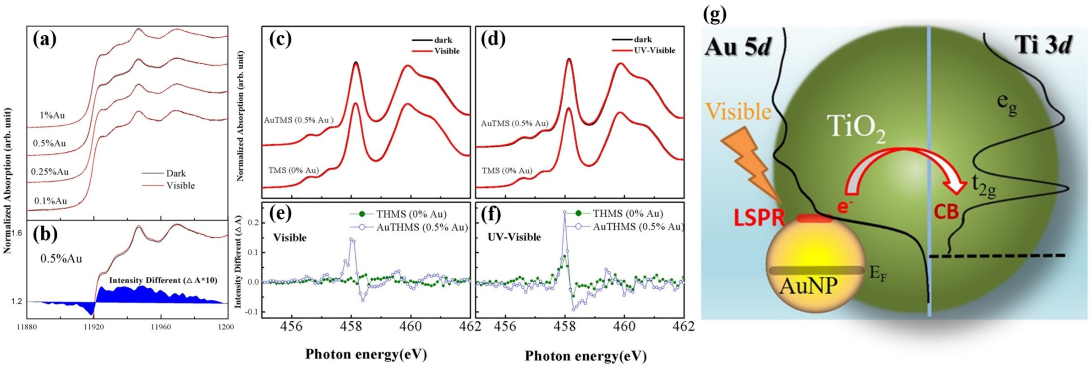


Figure 27. (a) Au L_3-edge XANES spectra of the AuTHMSs under dark and visible illumination. (b) Absorption intensity difference (ΔA*10) for the 0.5 % AuTHMSs. In situ XANES Ti L_3-edge for THMSs and AuTHMSs under (c) visible illuminated and dark and (d) UV-visible and dark spectra. (e,f) Difference in absorption intensity (ΔA) between samples under irradiation and dark from panels a and b, respectively. (g) Mechanism of plasmonic effect and charge transfer under UV-visible illumination in AuTHMS spheres. Reproduced with permission from Ref. [103].

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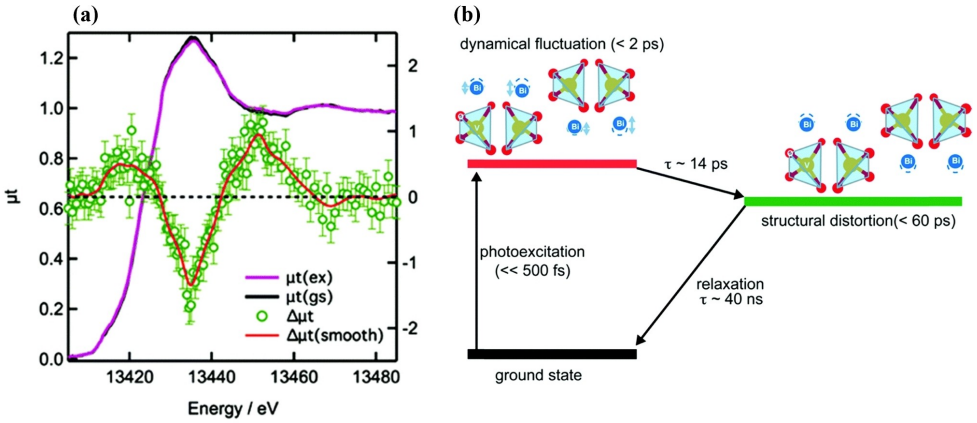


Figure 28. (a) Bi L_3 XANES spectra of the ground state (black), the excited state spectra at a time delay of 150 ps after photoexcitation (purple), and the corresponding difference spectrum (green circle) and its smoothed-line fit (red). (b) A proposed diagram of the photoexcitation process of $BiVO_4$. Reproduced with permission from Ref. [106].

orbitals contributed to the conduction bands for WO_3. Thus, photogenerated electrons could be photoexcited from an O 2p to a W 5d orbital in the photoexcitation process of WO_3, which could be explained by the W L_3-edge XANES spectra before and after the excitation in Figure 29.[107a] The electron in a W $2p_{3/2}$ orbital is excited to a W 5d empty orbital at the edge region, which gives a strong absorption peak known as a white line. Distinct changes of the XANES spectra are found around 10216 eV (Figure 29a). Especially, a blue shift was found at 150 ps after the excitation. Figure 29b shows plots of the white-line area and the valence state of W for reference compounds W, WO_2, and WO_3. From this plot, W^{5+} or the metastable WO_3 after photoirradiation was estimated as $40 \pm 5\%$ for the total number of W. The XAFS results could clarify the local structure change of W species, further explain the formation process for the metastable WO_3 through the structure of the metastable WO_3. Meanwhile, the distorted metastable structure could act as a reservoir of charge carriers which travel to the surface and to induce the reaction.[107c,d]

4. Summary and Future Outlook

In brief, this review summarized the recent research progress on the XAS applications in photocatalysis. As described above, XAS is a critical technique to provide abundant information of local electric and coordination structure and in-depth understanding of photocatalytic reaction mechanism. In particular, time-resolved XAS characterization based on synchrotron radiation with femtosecond time scale has been crucial in monitoring the changes in electronic and coordination structural properties during photocatalytic process. By virtue of the local structure sensitivity and high-resolution, time-resolved XAS based on synchrotron radiation has proven to be a valuable tool to address the scientific inquiries in kinetic processes that remain unanswered by other experimental methods under extreme conditions with the light illumination and high magnetic fields. However, photogenerated charge carriers' excitation and transport dynamics, transient states of photocatalyst, as well as the recombination mechanism of photogenerated charge carriers during the photocatalytic process are extremely challenging. Such time-resolved XAS measurements could explain the fundamental and transient phenomena occurring during photocatalytic reactions. It is

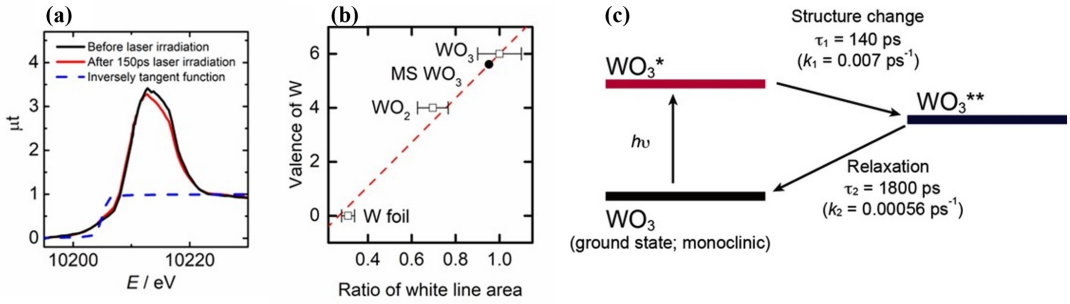


Figure 29. (a) W L_3-edge pump-probe XANES spectra of WO_3 before (black line) and after 150 ps irradiation (red line). (b) The relationship between the ratio of the area of absorption edge peak and the valence states of W. The black dot shows the result of the area ratio calculation from XANES spectra after laser irradiation, and (c) A proposed scheme for the photoexcitation process of WO_3. Reproduced with permission from Ref. [107d].

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believed that with the development of advanced synchrotron sources or X-ray free-electron laser (XFEL) with higher flux and better time-resolution, XAS spectra have more powerful applicability in discussing local characters including oxidation states, coordination structure and orbital hybridization on the time scale below the picoseconds (ps). On the other hand, radiation damage of samples may be a problem at such high fluxes and has to be considered very carefully. Recent XAS methodological developments are intended to improving classical XAS tests toward better control of XAS experimental parameters, such as time scale or probing depth. Moreover, until now, none of the characterization tools can comprehensively and deeply unravel the structure-property relationship of photocatalysts. Consequently, the combination of XAS with other advanced characterization techniques, such as aberration corrected scanning transmission electron microscopy (STEM), ambient pressure X-ray photoelectron spectroscopy (APXPS), in-situ FTIR, etc. should be given more attention. Through combining XAS with other analytical techniques from near-reality to extreme conditions, it allows scientists to understand the photocatalytic process in depth and guide the rational design of efficient photocatalyst.

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Conflict of Interests

The authors declare no conflict of interest.

Keywords: X-ray absorption spectroscopy · photocatalyst · synchrotron radiation · local structure · reactions mechanism

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